

LEARNING THE *Radiotelegraph* CODE

\$1.50

**BASIC
TRAINING
FOR
RADIO
OPERATING**



**For
INDIVIDUAL
OR
CLASSROOM
INSTRUCTION**



LEARNING THE RADIOTELEGRAPH CODE



**Published by THE AMERICAN
RADIO RELAY LEAGUE**

Newington, Connecticut 06111

U.S.A.

Copyright © 1970 by The American Radio Relay League, Inc.
Copyright secured under the Pan-American Convention. International Copyright secured. This work is Publication No. 18 of The Radio Amateur's Library, published by the League. All rights reserved. No part of this work may be reproduced in any form except by written permission of the publisher. All rights of translation are reserved. Printed in U.S.A.

Quedan reservados todos los derechos

Thirteenth Edition

Library of Congress Catalog Card Number: 55-8967

Foreword

THROUGHOUT THE NATION World War II brought great interest in learning the International Morse radiotelegraph code. Innumerable buzzer classes and radio courses sprung into being and thousands of people went industriously to work mastering the dots and dashes. Many requests were addressed to the American Radio Relay League, the national society of radio amateurs, to produce a piece of training literature that would be a reliable and helpful guide to those who were undertaking to learn this new language. It was to fill that need that this booklet was written. It did that job well.

Today its primary purpose is to furnish guidance in learning to send and receive the code, a requisite to the attainment of an amateur radio operator license, to those thousands who are newly taking up one of the world's most fascinating avocations.

It is from an amateur background of long familiarity and experience that there comes the ingenious idea here outlined for learning the code. The emphasis on learning the *sound* of letters rather than the sequence of their dot-and-dash components is strictly in accord with modern thought in the field. The applicability of this technique to code learning has been thoroughly tested; it is unquestionably the best way, as is also attested by its employment in the schools of the Army, Air Force and Navy.

There is something fundamental, almost universal, about this business of code. Its applicability far transcends radio and wire circuits. It is sent by blinker lights, by heliograph, by wig-wagging a flag. Automobile horns and the whistles of ships and locomotives can be "keyed" to make these signals, understandable to anyone who knows the code, and we have even heard an airplane pilot signaling a letter or two by manipulating his engine throttle. An unusual example, to be sure, but illustrating the thought that whenever there is any way of turning on and off for varying lengths of time anything that can be heard or seen, those who know this universal code have peculiar and often invaluable ability to communicate. So important is this knowledge, so frequent its benefits, that we have long thought that it should be imparted in the public schools, along with the ability to swim and to administer first aid.

You should know this code. It is the key to the first door of the fascinating hobby of amateur radio, the peacetime avocation of three hundred thousand enthusiasts. It is similarly a steppingstone to other radio fields. We hope this booklet gives you the help you need. Good luck!

JOHN HUNTOON
General Manager, ARRL

Newington, Connecticut

Contents

Learning the Code.	5
THE MECHANICS OF CODE—ELEMENT SOUNDS— LETTER SOUNDS	
Learning to Send	12
KEY ADJUSTMENT—FORMING CHARACTERS— ADVANCED PRACTICE—SPECIAL SYMBOLS—PRACTICE CIPHER GROUPS	
High-Speed Operation.	20
WORD SOUNDS—COPYING BY TYPEWRITER—"BUG" SENDING—AUTOMATIC EQUIPMENT—THE SUBCONSCIOUS MIND	
Operating on the Air	24
GENERAL OPERATING DATA—ABBREVIATIONS— MESSAGE FORM—LOG-KEEPING—RADIO OPERATOR LICENSES—CALL SIGNALS	
Code Practice.	31
TONE SOURCES FOR PRACTICE—A CODE INSTRUCTION TABLE—PRACTICE SCHEDULES	
Class Instruction	38
EXERCISES IN RECEIVING AND SENDING	

Learning the Code

- *The Mechanics of Code*
- *Element Sounds*
- *Letter Sounds*

YOU ARE about to learn a new language — the language of code.

The radiotelegraph code is essentially a language of sound. As such, it is quite difficult to illustrate in a book, and we shall have to ask that you follow each step closely and pay attention to every detail of learning as outlined in the paragraphs to follow.

One important preliminary: forget completely about "dots" and "dashes." Those terms were invented for the old land-line telegraph system, and they have absolutely no place in a modern system of learning radiotelegraph code. Forget, too, about any other code chart or table you may have seen or memorized.

The Continental (International Morse) Code is the one used in all radio work. It consists of various combinations of sounds and spaces, forming letters of the alphabet, numbers, punctuation marks and procedure symbols. The sound usually takes the form of a high-pitched audio tone having a frequency of about 500 cycles per second (approximately C above middle C on the piano), more or less, depending upon the receiving operator's individual preference. Lacking an audio oscillator itself for practice, the best way to simulate the high-pitched sound of code in headphones is to whistle.

THE MECHANICS OF CODE

To illustrate how closely allied are radio code and human speech, let us liken the mechanics of code to those of our own spoken English language. Human speech is made up of a number of pure sounds, such as "ah," "ee," "oh," and "oo" (as in "ooze"). We combine these to form other sounds, and those to form words. The sound of the letter I, for example, is not pure but it is made up of two others "ah" and "ee," spoken in rapid succession.

The radiotelegraph code is similarly constructed, but actually it is much easier to learn since there are but two fundamental sounds. In reality, each sound is the same tone and differs from the other only in length. The first is very short and staccato, the other somewhat longer (three times as long, actually). Since this high-pitched tone or whistling cannot very well be shown in print, for instruction purposes aural terms have been coined which closely duplicate the actual element sounds in code. We speak of the short sound as "dit," of the longer one as "dah." Combinations of these sounds, properly interspersed with spaces, are used to represent the various letters of the alphabet just as are the sounds in the spoken language.

For purposes of explanation to you and to whoever might assist you, let us say that the sound "dit" is pronounced as "it" with a "d" before it. The sound "dah" is pronounced with "ah" as in "father." The sound "dah" is always stressed or

accented — not in a different tone of voice, but slightly drawn out and the least bit louder. The sound "dit" is pronounced as rapidly and sharply as possible; for purposes of easy combination, as a prefix, it is often shortened to "di." When combinations of the sounds appear as one letter, say them smoothly but rapidly, remembering to make the sound "di" staccato, and allowing equal stress to fall on every dah. There should never be a space or hesitation between dits and dahs of the same letter.

These are simply convenient vocal terms to use in duplicating the sound of code. You should never think of a letter as being composed of certain separate sounds, but rather as a complete sound itself. You don't think of the spoken letter U, for example, as being composed of two separate and distinct sounds — yet actually it is made up of the pure sounds "ee" and "oo," spoken in rapid succession. You learned the spoken letter U as a sound unit itself. Similarly, you should learn code letters as sounds themselves, and not as combinations of other sounds. A skilled operator does not think of dits and dahs when copying, but actually hears the headphones speak words to him. He has mastered this new language. The sound "dah didididit dit" is just as familiar to him as the sound of the spoken word "the," for which it stands.

In no event should you use the "didah" table below — or any other such chart — for memorization study. If you do you will have a visual concept of code instead of an aural concept, and when you hear sounds you will have to convert them first into "didah" language, and then into letters. If you learn the sounds directly with their letters, however, without using visual references, you will eliminate this extra step while learning.

ELEMENT SOUNDS

Before taking up letter sounds, however, let us observe some of the element sounds. Practice saying to yourself the sound "didididit . . ." (dits in rapid succession). It should sound like a blast from a machine gun: staccato, evenly-spaced, precise. To make certain you get correct timing, start tapping the top of a table continuously in smooth, even sequence, like clockwork; if a metronome is available, it will serve admirably. The tapping (or metronome ticks) should be at about 100 per minute, or a bit less than two per second. Synchronize your tonguing of dits at four per metronome beat. Be careful not to say "didididit . . . didididit . . .", thereby leaving a space after every fourth dit; they should be as even and regular as the metronome beat, but four times as fast. If it will help, slightly accent the dit which coincides with the metronome beat. When you can do this easily, begin repeating "dahdahdahdah . . ." so that the beginning of each *second* dah synchronizes with a metronome tick (or table rap). Let the dahs run smoothly and make them of equal length. Do not let them become choppy; your voice should be almost continuous, broken only for that short instant your tongue cuts off the tone to make the "d" sound.

Now alternate, repeating dits for a moment and then switching to dahs, without stopping to take a breath. This will show you the proper proportion of dits to dahs at this speed, or any speed to which the metronome is set; that is, four dits or two dahs per metronome tick. You should now practice alternating this voice simulation with whistling. Again, make your whistled dits very short and staccato, your whistled dahs smooth and full.

A graphical representation of the relation of the metronome beat to the spoken dits and dahs might be that shown below; the beats are shown by small dots, and

Radiotelegraph CODE

the dahs are spelled "dahh" to complete the mechanical relation:

di di di di di di di di di di di di di di di di di di di
• • • • • •
dahhdahhdahhdahhdahhdahhdahhdahhdahhdahhdahhdahh
• • • • • •
di di di di di di di di di di di di di di di di di di di

Another excellent practice exercise is the alternating of single dits and dahs. Practice repeating (and then whistling) "didahdidahdidah . . .," and then "dahdidahdidahdi . . ." Here again the tone should be almost continuous, the dit as short as possible. An hour, in several shorter periods, is not too much time for these exercises. If you can master them, you will have no difficulty in forming the various letter and number sounds.

LETTER SOUNDS

When you have mastered the timing and rhythm exercises explained above, you can begin to combine these groups and form letter sounds. In doing so, remember that this is a new type of spoken language you are learning, not a collection of short and long code symbols taken from a printed page. In fact, it might be better to forget that you are learning "code" and think of it instead as the "didah" language.

This is how you should proceed to learn the letter sounds: if possible, get someone who knows the code to start you off. Or, a member of your family or a friend (preferably not one also wishing to learn code) will suffice if he will study these paragraphs and practice the subsequent voice exercises a short time. Give *him* the book. Then ask him to pronounce the sounds to you, identifying them with the letters for which they stand. Take a few letters at a time, such as in the groups suggested below. As your "instructor" sings out the sounds to you, you should be able to call out the letters, or vice versa. This is excellent practice. It will come harder at first, possibly, than other ways, but you are set for a successful code career if you learn by sound. In other words, after you read this section do not go back and memorize the code tables. Don't even take another look. Get someone to call out the sounds to you, either by "didah" language or whistling, and preferably a combination of both.

Let us take a few letters and numbers to start with, as shown in the following table. Remember that you should look at this table only briefly while reading this text; when you come to study, have someone hold this book and coach you in memorizing the code sound of each letter.

5	didididit
0 (zero)	<u>dahdahdahdah</u>
E	dit
T	<u>dah</u>
A	<u>didah</u>
R	<u>didahdit</u>

LEARNING THE

The "5" should merely be five *staccato* dits in the same sequence and speed you practiced earlier; the zero should similarly be five dahs. "A" is our first character using both sounds. Make that di very short, the dah the usual length, properly stressed; it should have the same metric swing as does the word "to-day," rapidly spoken and strongly accenting "day." Similarly, the "didahdit" of "R" should have the same metric swing as "repeated," the second syllable strongly stressed, the final one not accented and as short as you can make it.

Spend at least one half hour on this group and preferably more, although your practice should then be split up into shorter periods. Take your time and learn the sounds of these letters thoroughly. Repeat them in "didah" language, and then whistle the characters. You will find it interesting to make up some words from this group of letters, such as:

ATE RAT TEAR 5Ø ERA RATE Ø5 ART TARE

You should immediately begin practicing copying down the characters your instructor calls out to you in sound. Copy them simultaneously, speaking the letter if you like. For the first copying practice it is well to print; in the transmission of non-English text, which you often will be getting for letter practice, longhand letters are sometimes confused with each other. For example, L is mistaken for E, an uncrossed T for L, etc. The Army Signal Corps requires printing ability by its field operators, since much of the text handled is in code groups and errors would cause much difficulty in reading the actual message. Print until you are able to receive about ten words per minute, above which speed you will want to resort to longhand to keep up with the text.

Learn to print the characters rapidly and without conscious effort; your mental effort must be on reading the code sounds and not on your finger movements.

5 Ø E T A R	W K Z M
S L U Q J	D X F G
H O N C V	↓ 2 3 4
↓ B Y P	6 7 8 9

Signal Corps operators learn this method of handprinting letters, shown above in the groups by which you will learn letter sounds.

Never look back over your copy when receiving, nor try to guess what word is coming; copy what is sent. If you do not immediately recognize a character, skip it and devote your full attention to the next one; if you try to think back, you doubtless will miss several letters in a row. By checking your written copy with the transmitted text, you can determine what letters are giving you trouble and give special attention to them.

Radiotelegraph CODE

Here is another group of letter sounds. You are ready to learn them when you can *instantly* call out English letters for code sounds spoken to you, or vice versa. Before beginning their study, have your instructor review you on the first group; do not be so eager to learn fast that you forget old sounds! Intersperse learned sounds with new ones when studying this next group:

S	dididit
L	didahdidit
U	dididah
Q	dahdahdidah
J	didahdahdah

The first letter obviously is simply three staccato dits. L is more difficult; its metric swing is like that of "fraternity"; again, make dits rapidly, particularly the final ones, and stress the dah. U is similar to S with the final dit changed to a dah; practice the two letters interchangeably to get the rhythm. Q and J should be smooth; stress each dah equally, make the dit short.

Study these in the same manner as the previous group. You have nearly twice as many letters to remember now, so your progress will be a bit slower. Don't rush; learn each sound thoroughly before proceeding to the next. Speed will come later, and it will come rapidly later if you learn by sound. There are quite a number of words which can be made from the letters so far studied, more examples of which are here given:

SELL REQUEST JAR LURE RUSE STARE
SUET SQUEAL JET QUEER SLATE JUTE
QUART LESSER QUELL JUST TESLA SALES

Practice saying the sounds to yourself, particularly between study sessions. Occasionally you should have an operator who knows code check you on your progress.

Here is a third group, to be taken up only after you have learned the previous eleven characters well:

H	didididit
O	dahdahdah
N	dahdit
C	dahdidahdit
V	didididah

LEARNING THE

The first two letters should give no trouble. Be careful not to have a space in the sound for N; nor to make the dit anything but short. C can best be simulated by remembering our earlier exercise of alternating single dits and dahs; practice that again momentarily, and you should have no trouble.

When learning the sound of V, practise H and then make the final dit a dah; practice the two letters intermittently as you did U and S, repeating in your mind what letter is concerned with each sound. Again, some suggested practice words for the new letters:

COVERLET ALONE CANCEL VOCAL HOVER THE

COLLECT NEVER THAT SHONE LATHE SHEET

THESE CONQUEST THERE NEUTRAL SEVERE

TENET RUNNER ENCLOSE RELUCTANT JOCLAR

JOVE HELEN LUNAR CONCLAVE RECLUSE

Here is a fourth group of letter sounds

I	didit
B	<u>dah</u> dididit
Y	<u>dah</u> didah <u>dah</u>
P	didah <u>dah</u> dit

By now you should have developed sufficient timing sense and code consciousness that you can pronounce new sounds without difficulty. Simply remember to keep characters smooth, without spaces.

In between practice periods, when convenient, notice street signs and posters containing these letters; when you see one you know, call out its code sound. Keep this practice principle always in mind and use it a great deal; you will find a goodly amount of idle time you can put to good use — riding back and forth to work, or walking to the corner drugstore. Or even if you are not in the vicinity of signs, there are a number of short words you can practice saying to yourself in sound language. Make your character formation snappy; leave a recognizable space between letters, plenty of space between words. Use both whistle and voice technique. We want to get you thinking subconsciously of code whenever you see letters and words. This spare-time practice is probably the most important single factor in the progress you make.

Try some words like these to gain familiarity with the new letters:

BOTTLE CORNCOB YESTERYEAR HONEY THIS

POPPY JITTERS BATTERY BAY SISSY

RECONCILIATE COUNCIL BONY PHONE JOIN

SUPPER PAVE SHIP PAY CAPON NYLON COIN

PYRITES COPY BOIL PEBBLE CAVITY VICIOUS

Radiotelegraph CODE

Now, another group:

W	<u>didahdah</u>
K	<u>dahdidah</u>
Z	<u>dahdahdidit</u>
M	<u>dahdah</u>

When these have been thoroughly learned, you may proceed to the final group:

D	<u>dahdidit</u>
X	<u>dahdididah</u>
F	<u>dididahdit</u>
G	<u>dahdahdit</u>

Here are some practice words for the last two groups:

WORK BUZZER ZERO SLOW HARK WARMER
JERK SUZANNA BUMP QUIRK MAKE WINK SIMMER
MAP SKIM MILK TOMORROW WHOLESOME JAM
QUALMS LOWLY WHARVES POMPON WAR MARK

DOCK KIND FINGER DOLORES DEXTEROUS GOLF
KIDDER FIXTURES GOODLY JIGGER JINX FOGGY
STUFFING DOG FLIGHT DRAUGHT FLING FOX GOD
FADDIST GUFFAW DAGGER MEXICAN DODDERING

You already have learned the two simplest numerals: five and zero. The others are:

1	<u>didahdahdah</u>
2	<u>dididahdah</u>
3	<u>didididah</u>
4	<u>didididid</u>
6	<u>dahdididit</u>
7	<u>dahdahdidit</u>
8	<u>dahdahdahdit</u>
9	<u>dahdahdahdahdit</u>

Learning to Send

- *Key Adjustment*
- *Forming Characters*
- *Advanced Practice*
- *Special Symbols*
- *Practice Cipher Groups*

IT is important that you should learn the correct sound of code letters thoroughly before ever touching a telegraph key. If you do not know how a code letter should sound, no amount of playing with a key will teach you. When you are at the point where you can unhesitatingly call out each letter as your instructor pronounces or whistles the sound, you are ready to learn how to handle a key. First, however, you need some device to produce a tone. A buzzer set will serve the purpose but a much better tone source is the vacuum-tube oscillator, since it duplicates the audio-frequency beat note a radio operator hears. Construction data on both types are given later in this booklet.

KEY ADJUSTMENT

A telegraph key is simply an on-off switch in convenient form for rapid manipulation. Pressing the key knob closes the electrical circuit and produces the "mark" or sound; releasing it allows the contact to break, producing the space or no-signal period. Correct key adjustment is that adjustment which fits your particular touch, and it is important that you arrive at the correct adjustment before attempting to use the key. A reliable rule is that there should be a vertical movement of about one-sixteenth inch at the key knob. This is measured from the top surface of the knob. It is set by the rear screw adjustment. When the knob's top surface goes down about one-sixteenth inch upon pressing the key, you have the approximate "average" spacing between the key contacts. In making any key adjustment, be sure to loosen the lock nuts first so that you do not strip the threads. Tighten all lock nuts when you complete the adjustment. The contact points should be perfectly aligned by means of the side screws. There should be a very slight degree of side play between the two side screws. These screws should be tightened, then loosened just a bit, so that the key moves freely and does not bind. Recheck the contacts after this adjustment to be sure they are true, making any slight readjustment necessary.

Although the "one-sixteenth inch" rule is a good one to follow for first adjustment,

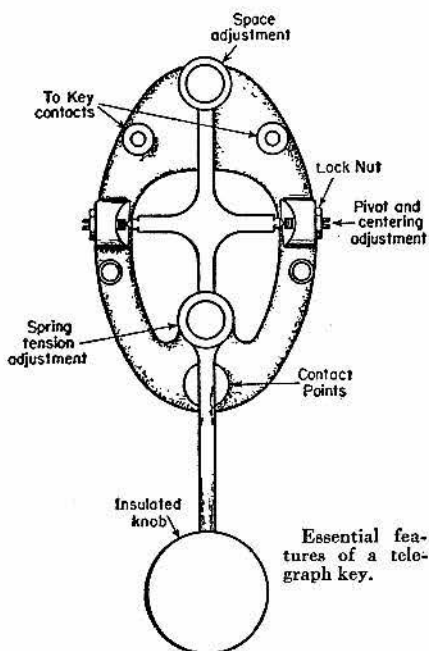


Illustrating the correct position of the hand and fingers for the operation of a telegraph key.

the amount of vertical movement can later be changed to suit your particular "fist." The spring tension, likewise, must be set for the individual operator. Some prefer a heavier spring than others. The primary consideration is to send *good code*; how you have your key adjusted to do this depends on what you find best for you. However, it should be remembered that too heavy a spring tends to make your sending "choppy," causing you to "clip" your dahs and dits, as well as being tiring for long periods of sending. Similarly, too light a spring tends to cause you to run characters together, there being insufficient control of the key. Remember, *you* are making the characters, the key isn't! Generally speaking, a somewhat heavy spring allows better control, particularly of dits. With a spring of "feather-weight" tension, the dits are likely to run away from you and you will find yourself slurring them. No treatise on key adjustment can ever solve the individual's problems in this line. Only by personal use can you find the correct adjustment for yourself.

There is a definite sending posture which should be observed. Sit upright in your chair, square with the operating table, with your arm on a line with the key. The key knob should be about eighteen inches from the front edge of the operating table, allowing room for the elbow to rest on the table. The muscle of the forearm should support the weight, and the wrist should be off the table. A table about thirty inches in height is best. The key may be fastened by means of wood screws directly to the table if one is available for permanent use. Otherwise, it may be fastened to a rectangular piece of thin board such as three-ply veneer, about six inches wide and two feet long. The manner of grasping a key knob is also the choice of the individual operator. When learning, place the thumb against the left edge of the key knob, the first finger on top of the knob at the rear and lapping over the rear edge just a bit, and the second finger against the right edge of the knob, about in the center or slightly to the rear of center. In no event should the grip be tense. The first and second fingers should be slightly arched, not held out straight. The third and fourth fingers should be permitted to curl naturally toward the palm of the hand, but they should not be tightly clenched. Keep the fingers, hand and wrist relaxed at all times.

Now that you are all set, you can begin sending practice. Before taking up letter sounds it is best to achieve smoothness and facility with element sounds, as we did in learning by voice. Begin by making a series of dits — ten or more in a row. Make them evenly spaced, precise, a bit slower than the speed at which you learned the voice sounds. Work on both dits and dahs, and then alternate, just as you did in voice work. Here again a metronome, or a substitute such as someone rapping on the table for you, will be a valuable timing guide. Remember to synchronize your sending at four dits, or two dahs, per beat. Make the beats about 60 per minute —



LEARNING THE

— one per second — as compared to the 100 in receiving practice. Synchronize your sending with the spoken sounds, if you wish. Keep your wrist flexible; allow it to bob up and down with your sending. If you find that your fingers or whole arm are doing the work, stop, and start over again. Be particularly careful to make dahs smooth and full; the tone should be almost continuous, broken only for that tiny instant the key contacts open.

FORMING CHARACTERS

Once the rhythms of dits and dahs have been mastered, we can proceed to the consideration of the correct formation of each character. This is an important consideration, for if we do not make every letter and numeral correctly we shall never send good code. There can be no characters sent incorrectly. It is one thing to know how we want a character to sound but another thing to make it sound that way! Let's analyze the correct manner in which to make each character. Starting with the basic letters E and T, practically every character is composed of combinations of other characters. A study of the correct rhythm for each character (how it sounds when sent correctly) discloses that certain groups of letters and numerals are associated. It is urged that you practice the characters in accordance with the following plan. The letters listed in each group are more or less associated and it is recommended that practice be "by groups," with each letter mastered in the order given within each group. Keep your speed at the level at which you make a minimum number of errors. Accuracy and perfection come first — speed will come with practice. Here again, it is well to have someone skilled in code check your progress occasionally.

Group No. 1: The characters E, I, S, H and 5. With the correct dit rhythm mastered, you can send all of these dit characters. If you experience any difficulty, go back and send a series of evenly-spaced dits (didididididit, etc.). When your wrist responds, send a 5, using the same rhythm as for the longer series . . . send an H likewise, etc. This same dit rhythm carries over into other characters which are made up of didah combinations.

Group No. 2: The characters T, M, O and 0. These were covered under the subject of correct dah rhythm. Little difficulty is usually encountered in making dah characters. If, however, you find yourself sending them jerkily, go back and practice the dah series exercise (dahdahdahdahdahdahdah, etc.) until you regain the feel of this action.

Group No. 3: The characters A, R, L, W, J, I and P. These all have as their groundwork the letter A (didah). First master that letter by sending several times a smooth didah; then send an R (didahdit), which is actually the letters AE sent together as one character. Always bear in mind that the space between any parts of a character is the same length as a dit. Next try an L (didahdidit), which is actually AI sent as one character. Next send W (didahdah), thinking of it as AT sent together. If you can send A successfully, you can send R, L and W similarly. The letter J (didahdahdah) should be thought of as WT sent together. The numeral 1 (didahdahdahdah) should be considered as JT sent as a single character. It will be noticed that these combinations give a smooth continuity for each character. The letter P (didahdahdit) should be considered as WE sent as one smooth character. Care must be taken in sending each of these characters that no additional

Radiotelegraph CODE

space is left between the two letters making up the complete character. For example: although P is WE, take care that you do not leave more than one dit length between the W and E. The overscore indicates that the two letters are sent rhythmically as one sound (didahdahdit).

Group No. 4: The characters U, F, 2, V, 3 and 4. Master first the letter U (dididah). Do not think of U as any combination of letters; think of it only as dididah sent smoothly. (A dangerous misconception is to think of U as the letters IT; avoid this, or your character is apt to sound jerky.) Think only of the sound dididah. Next, tackle the letter F (dididahdit), using the basic letter U as a foundation. F is actually UE sent as one character. If you can send U, you can send F. The numeral 2 (dididahdahdah) is made up of the letters UM sent together; think of it this way and you will get a smooth 2. The letter V (didididah) must not be considered as any combination of letters. (A common fault is to consider V as ST, resulting in a jerky character.) Think of V only as the sound combination didididah. Practice this, using the dit rhythm, until you send a smooth letter V. One of the most difficult characters for many operators is the numeral 3 (didididahdah); it is too often sent jerkily (as SM). To send a smooth numeral 3, think of it as VT sent together; there you will have the smooth continuity of a correct 3. Conquer the letter V and you will have no trouble making 3. The numeral 4 must be considered only as the sound combination didididah. Do not attempt to consider the 4 as composed of any letter combinations (as HT) or you will get a jerky character. In sending the 4, use the same dit rhythm as you do for the 5, making the last unit a dah instead of a dit (didididah).

Group No. 5: The characters N, D, B, 6, 8, 9 and X. The N, of course, is merely a reversal of A. Practice the N (dahdit) by sending several times a smooth dahdit. In making a D (dahdidit), avoid making it sound jerky (as TI). Think of it only as the combination of sounds dahdidit, sent evenly. Likewise, the letter B (dahdididit) must be thought of only as the combination dahdididit. (It is a mistake to think of B as a letter combination, such as TS; this results in a jerky character.) The numeral 6 calls for the same rhythm as a numeral 5, except that our first unit is a dah rather than a dit. Think of 6 only as the smooth sound combination dahdididit, evenly spaced. The numeral 8 (dahdahdahdidit) must not be considered as the combination OI; this results in jerkiness. It is best to think of it only as the rhythmic combination dahdahdahdidit. However, the combination MD, with care taken not to leave additional space between M and D, will give you a correct 8. The numeral 9 (dahdahdahdahdit) calls for the same dah rhythm as does the numeral 6, with the exception that the last unit is a dit instead of a dah. The letter X (dahdididah) should be considered only as the combination dahdididah. (Avoid thinking of it as a letter combination, such as DT or TU; these tend to result in a jerky character.) In practicing X, think only of the rhythmic dahdididah, dahdididah, etc.

Group No. 6: The characters G, Q, Z, 7, K, C and Y. Consider the letter G (dahdahdit) as the sound combination dahdahdit. With care to keep the spacing correct it might be thought of as ME sent together, but it should never be thought of as TN. (This usually results in an uneven character.) Next, send a Q (dahdahdidah), considering it as MA sent as one character. (Avoid thinking of Q as GT, which makes for jerkiness.) The letter Z (dahdahdidit) may be considered as TD sent together smoothly, although it is better to consider it only as the sound dahdahdidit.

LEARNING THE

The numeral 7 (dahdahdididit) can safely be considered only as the sound combination dahdahdididit, sent evenly. (A common mistake is to think of 7 as MS, a jerky combination.) The letter K (dahdidah) must be thought of only as the sound dahdidah, smoothly executed. (Do not think of K as any letter combinations, as TA or NT, both resulting in unevenness.)

Probably the most troublesome of all letters is C (dahdidahdit). In order to get a smooth character C it is recommended that it be thought of as the combination KE sent as one character. Here you will find a smooth rhythm, as opposed to the combinations TR or NN, which are almost sure to result in jerkiness. In actual practice many operators have found the KE combination their answer to correct formation for the letter C. Similarly, the letter Y (dahdidahdah) should be considered as KT, sent together smoothly, as opposed to TW or NM. KT will give you a rhythmic Y. Master the letter K and you will have no trouble with C or Y.

TABLE OF CORRECT SOUND COMBINATIONS FOR SENDING PRACTICE

Consider each letter as its individual rhythmic sound combination. If you have difficulty forming certain letters to make them sound right try the letter combinations suggested, mastering the first letter, then adding the second unit, sending the whole as one character.

- A — di-dah (not ET)
 - B — dah-di-di-dit (not TS)
 - C — dah-di-dah-dit, or KE (not TR or NN)
 - D — dah-di-dit (not TI)
 - E — dit
 - F — di-di-dah-dit, or UE (not IN)
 - G — dah-dah-dit, or ME (not TN or TTE)
 - H — di-di-di-dit
 - I — di-dit
 - J — di-dah-dah-dah, or WT (not EO or AM)
 - K — dah-di-dah (not TA or NT)
 - L — di-dah-di-dit, or AI (not ED)
 - M — dah-dah
 - N — dah-dit (not TE)
 - O — dah-dah-dah (not TTT)
 - P — di-dah-dah-dit, or WE (not EG or AN)
 - Q — dah-dah-di-dah, or MA (not GT)
 - R — di-dah-dit, or AE (not EN)
 - S — di-di-dit
 - T — dah
 - U — di-di-dah (not IT)
 - V — di-di-di-dah (not ST)
 - W — di-dah-dah, or AT (not EM)
 - X — dah-di-di-dah (not DT or TU)
 - Y — dah-di-dah-dah, or KT (not TW or NM)
 - Z — dah-dah-di-dit, or MI
-
- 1 — di-dah-dah-dah-dah, or JT
 - 2 — di-di-dah-dah-dah, or UM (not IO)
 - 3 — di-di-di-dah-dah, or VT (not SM)
 - 4 — di-di-di-di-dah
 - 5 — di-di-di-di-dit
 - 6 — dah-di-di-di-dit (not TH)
 - 7 — dah-dah-di-di-dit (not MS)
 - 8 — dah-dah-dah-di-dit, or MD (not OI)
 - 9 — dah-dah-dah-dah-dit
 - 0 — dah-dah-dah-dah-dah

Radiotelegraph CODE

In all of these examples it should be emphasized that the overscored letter combinations represent one sound combination, not the *two* sounds of each individual letter concerned. For example: $\overline{KT}$ stands for dahdidahdah (Y), one rhythmic sound; *not* dahdidah space dah, two separate letter sounds. In practicing all characters, repeat the sound of them aloud to yourself. (Example: In practicing B, repeat to yourself dahdididit, dahdididit, dahdididit, etc.)

Determine your difficult characters and practice them until you master them. With the correct formation of each character as a basis we can proceed to combine them into words, giving consideration to correct spacing between characters and between words. With incorrect formation of even one character, our sending will lack that "something" which distinguishes between the perfect sender and the run-of-the-mill. That is why we must first consider each character separately and practice each separately until we master them all. Then — and then only — should we start sending words.

ADVANCED PRACTICE

To become expert in transmitting good code, after you have thoroughly learned each letter and numeral and can both send and copy letters without hesitation, your best practice is to listen to commercial automatic tape stations. Perfectly-sent code can be accomplished only by a machine, and you want to get fixed in your mind, indelibly, the correct formation of each and every code character and, in particular, the associated spaces. Notice the formation of each letter, the spaces left between letters and words, and the proportion in length of dits to dahs.

One of the best methods for deriving this association is to find a commercial or other tape station sending at about your maximum receiving speed. A few schedules of press transmissions which will be helpful in practice are given on a later page. Listen to the transmissions as you would a musical concert, concentrating on assimilating every detail.

The spaces between words may seem exaggerated, simply because you have probably been running yours together. A score of other details where the automatic transmission is different from yours will very likely show up in the same text. From all this you will learn where your own faults lie and be able to correct them.

If you can locate a tape station sending double (repeating each word), you can get excellent rhythm practice. Set up an audio oscillator alongside your short-wave receiver so you can hear both simultaneously. As each word comes through the receiver fix it in your mind; then, as the tape repeats it, send the same word on your oscillator simultaneously with the tape, as closely to perfect synchronism as possible. Perhaps you will find yourself leaving too much or too little space between characters,

DIT	■	1	} TIME DURATION RATIO, RELATIVE TO "DIT" AS 1
DAH	■	3	
ELEMENT SPACE	□	1	
CHARACTER SPACE	□	3	
WORD SPACE	□	7	



This chart shows the mechanical relations between code element sounds, letter sounds and word sounds.

LEARNING THE

or making certain dahs too long — these are the most common errors. Remember that all inaccuracies are yours, and profit accordingly. By such constant practice you will learn the proper rhythm and precision of perfect code. It's bound to work itself, subconsciously, into your sending.

Probably the most important single factor in sending ability is this sense of proper spacing. While it can be much more easily obtained by listening to tape than by visual study, the student should understand the mechanical relation of the various marks and spaces, as shown in the chart on page 17.

SPECIAL SYMBOLS

In transmitting text there is, of course, a need for code symbols for punctuation marks and some special procedure signals to facilitate rapid transmission. The important ones are:

Period	<u>didahdidahdidah</u>
Comma	<u>dahdahdidahdah</u>
Question mark	didah <u>dah</u> dit
Double dash	<u>dah</u> didid <u>dah</u>
End of message	didah <u>didah</u> dit
End of work	dididid <u>dah</u> dah
Wait	didahdididit
Invitation to transmit	<u>dah</u> didah
Hyphen	<u>dah</u> didididid <u>dah</u>
Parentheses	<u>dah</u> didah <u>dah</u> didah
Colon	<u>dah</u> <u>dah</u> <u>dah</u> dididit
Semicolon	<u>dah</u> didahdidahdit
Quotes	didahdidid <u>dah</u> dit
Error	dididididididit
Apostrophe	didab <u>dah</u> <u>dah</u> <u>dah</u> dit
Fraction bar	<u>dah</u> didid <u>dah</u> dit
Understood	dididid <u>dah</u> dit
Attention	<u>dah</u> didahdidah

Radiotelegraph CODE

PRACTICE CIPHER GROUPS

Cipher groups make better practice material than plain English because the student can't foresee the next letter. The groups below, taken from the operator's manual of the Signal Corps, are representative both of the practice material used in military schools and of ciphered messages actually used in the services. Their breakdown into 5-letter units makes it easy for you to determine your approximate speed, figured on the basis of 5 characters to a word.

With someone to send to you, this material of course provides the best possible copying practice. Check back for your errors, concentrate on the letters that are proving difficult for you. If you find yourself memorizing some of the combinations or their order, use each group backwards or start from the bottom of the page. There are plenty of practice possibilities in the following list, even for the experienced operator who is seeking higher speed on the typewriter.

O K I C Q	A E I O U	M V H V D	J E H D Y	Z M N Z B
2 4 6 8 Ø	1 3 5 7 9	A P L K M	M C N C B	N D B G Y
C K T O G	C J R N O	N D H B H	P A O K U	K D I O E
A X B T R	Y O U M K	J T I O E	M D H N H	C V Q A R

Ø 3 7 5 9	C E M I D	M N L F G	U E Y T A	N A S T F
C W X C K	J X E F Y	Y O U M K	Q Y E U R	C F G I L
D K A K X	E F M E Y	V O I U M	Q W Z X N	C A D X A
W Q Y F Z	4 9 2 8 5	1 2 7 Ø 9	M N O P A	T T L O R

O M I W G	J N A B D	9 7 2 Ø 1	K Q Z A X	M A I D P
H S P C Q	J B C Y D	E F X N K	L A R D O	7 6 3 2 1
X G L D T	P G Z U N	L C K Q H	M Q E C V	7 Ø 3 6 5
8 Ø 7 Ø 1	C T I L L	O C Y A A	B E X Z B	O H W I M

R K M O Z	T H Q V I	Z L J B I	A L C R J	V T N E E
E K J C D	H M O U S	8 4 7 Ø 7	D L Z I K	H E L K F
D J H N G	T V C P T	K D K N G	P H V T X	4 7 3 8 2
H H H G J	C P T K O	F H F U X	H I M A D	M R F X E

L D O S J	2 Ø 1 8 4	N T V M G	C P N Z I	4 Ø 3 6 7
1 2 3 1 Ø	V N F B H	W O H Z R	J B O A H	O B R D K
B A Y Q T	D I E U Y	I P A Q Z	G Y E V Z	P V F K Q
Q N B A V	J C U W H	F N Z Q J	A R J Z U	R B E F P

2 5 8 9 4	8 3 7 4 6	P G M A E	D X A Q N	K D I E Y
J X H G Q	L K D O P	1 Ø 2 7 3	1 1 9 Ø 2	M D I D H
9 8 Ø 1 5	W E Q F Q	D X N Z E	Ø 6 1 4 3	Ø 9 1 6 5
M B U D F	1 7 8 6 3	9 8 2 3 4	C P T O K	Q Z A S U

High-Speed Operation

- *Word Sounds*
- *Copying by Typewriter*
- *"Bug" Sending*
- *Automatic Equipment*
- *The Subconscious Mind*

In high-speed copying a new principle is involved. It is one you will reach automatically if you progress sufficiently far in your practice, but it is worth detailed explanation here.

When one first learns the code by sound, he learns letters first. (In some cases, students studying visually learn *parts* of letters first, but we have tried to obviate that error by teaching code sounds in letter-units.) With a good deal of practice, one may slowly increase his speed of copying until he reaches a certain point — differing with various individuals — which is the maximum speed at which he can copy individual letters without having them seem to run together or blur in his mental thought. The average is around 28 words per minute.

WORD SOUNDS

Progress beyond that point must be on a new principle of copying, then. It is simply the process of copying by *word* sounds instead of letter sounds. An operator capable of receiving, say, 25 words per minute, can listen to 35 or 40 w.p.m. text and easily pick out the shorter, more common words, such as "the," "and," "but," and so on. He can do so only because he is copying *word* sounds and not letter sounds. A skilled operator does not hear letters but actual syllables and words. The code sound "didit dahdit dahdahdit" (the familiar suffix "ing"), for example, is mentally heard by the skilled operator as a complete sound and not as three different letters; again, it is as if someone had pronounced the syllable to him. When you reach this point, you appreciate that the radiotelegraph code is, as we said at the start of this booklet, truly another language.

However, there is no secret to the attainment of this ability, except continued practice. You can help yourself, though, by having someone send to you rapidly (about 10 w.p.m. above your normal receiving speed) the common words and syllables such as *and*, *of*, *the*, and so on. However, nothing can equal the practice obtainable by copying commercial tape transmissions.

It should be pointed out that one important prerequisite to high-speed copying is ability to spell. Since you do not hear letters, but entire word sounds, spelling of a word on the typewriter is up to you. True, it was sent only one way, but since you did not hear individual letters you do not know exactly which were sent; you know only what the word sounded like to you. You must know the spelling of words before you can recognize them from their code sounds and be able to transcribe them correctly to the typewritten page.

COPYING BY TYPEWRITER

The mark of a good receiving operator is ability to make perfect copy at high speeds on the typewriter. Ability to read rapid code "in your head" means little; what counts is what you can transcribe to paper correctly. Since the limit of hand-writing ability is about 30 words per minute, one must resort to a "mill" for copy at higher speeds. This necessitates skill as a touch typist; few "hunt-and-peck" typists ever became really good operators.

These common English words and syllables, if sent to you by a good operator at high speeds, make excellent practice for acquiring the ability to grasp word sounds in code..

THE	ONS	ITS	THAT	WHICH
ING	ESS	WHO	HAVE	THEIR
AND	AVE	HIM	WITH	THERE
ION	PER	MAY	FROM	WOULD
ENT	BUT	OUT	WERE	THESE
HER	NOT	OUR	THEY	SHALL
TER	DAY	ANT	WILL	GREAT
COM	INE	NOW	THIS	OTHER
ALL	ONE	YET	WHAT	ABOUT
CON	SUB	TWO	MORE	THOSE
MAN	WAS	OWN	UPON	COULD
EST	YOU	OLD	LIKE	FIRST
FOR	HIS	NEW	SUCH	MIGHT
ATE	HER	CAN	INTO	AFTER
ERS	SHE	ANY	TIME	TRANS
NCE	HAS	TED	VERY	RADIO

The first step, then, is to acquire that ability — by home study in a standard course, getting someone to tutor you if possible, or by attendance at some school such as an evening public school. Your typing should be nearly faultless in accuracy before you attempt code copy, although a high speed is not necessary immediately. For practice, your typewriter should be placed on a table separate from that supporting the code-practice oscillator or receiver, as the case may be. A portable typewriter table, on roller casters, is excellent. Sponge-rubber pads under the typewriter feet will help eliminate vibration.

Use standard letter-size paper, and double space between lines. Do not try to capitalize any letters at first; all lower-case type will suffice until you become proficient. Do not attempt to copy from a loudspeaker, for even with the so-called noiseless variety of "mill" its racket will be interference to the code sounds. Use 'phones. For sustained periods of practice, a set of sponge-rubber earphone pads will be found comfortable. If you find the headset cord getting in your way, bring the leads off the back instead of the front of your body.

Pick a station on your receiver sending at a speed you can comfortably copy or, if you are using a code machine, set its speed accordingly. Do not expect your first hour or so of practice to be encouraging; you have to learn coordination between typing and the mental processes of copying, and that takes time. It is like the old trick of trying to pat one's head and rub one's stomach simultaneously — it's not so easy as it sounds.

In practicing, do not listen and then type ferociously for a second . . . and listen . . . and type hurriedly again. Your typing must be disassociated, consciously, from your code reception. After you are able to handle this first speed, pick stations sending a bit faster (or step up the code machine), so that you get about 90 per cent of the text; when you copy solid, again step up the speed.

"BUG" SENDING

The complement of high-speed receiving is of course high-speed sending. You should never let your sending speed outstrip your receiving speed. However, if that does happen, it simply means that your conception of the code is mechanical rather than in terms of sound. A *semiautomatic* key, or "*bug*," can be used for speeds higher than obtainable on a "straight" key, but in no event should a student attempt to handle a bug until he has mastered the regular key. The semiautomatic key is a mechanical device which produces a series of dits (when the proper lever is tripped) by means of a vibrating contact. Dahs are made in the usual manner, although the operating knobs work in horizontal instead of a vertical motion. Relieved of the fatigue of tense motions involved in making rapid dits, the fingers are able to tap out code of much higher speed.

The position of the bug should be similar to that of the straight key. The arm again should be relaxed, and the right side of the hand should rest on the table immediately in front of the key. To the left of the control paddle is the thumb, which when moved to the right trips the dit vibrator; to the right of the paddle are the index and second fingers (some operators prefer to use only one), which when moved to the left operate the dah contact, similar to a straight key set on edge.

Operating motions consist of an easy roll of the wrist and hand from one side to the other. With the key connected to a tone source, practice sending a series of dits, then of dahs, then of alternating single sounds — just as we did in learning voice sounds. Take letter sounds in that sequence, as well. Before attempting to use a bug in actual operating, you should become proficient with it through practice with an audio oscillator and headphones.

An "electronic key" goes a step further by generating automatic dahs as well as dits, through electronic circuitry rather than by vibrating mechanical contacts. As the proper dit-dah-space relationships are fixed within the unit, a capable operator can send code with extreme precision and with minimum effort. For the same reason, an inexperienced operator with a tendency to form characters with less than machine precision will produce gibberish from the unit's output. Accordingly, it is especially important to practice at length with an electronic key before actually using it in communication.

THE SUBCONSCIOUS MIND

A person just learning to drive a car has no time for conversation; his every mental and physical effort is bent toward operating the car. Yet after a few months' experience he can carry on a conversation, occasionally watch the scenery, smoke a cigarette, etc. — because he has learned to drive automatically, subconsciously. A good high-speed operator reacts instinctively, subconsciously, to incoming code — just as a good typist automatically taps out words without thinking of the position of the individual letter keys.

This may hurt, but the reader may as well know the truth: the ability is accomplished principally by practice and more practice. Think in code, talk in code, visualize code in place of street signs; make it second-nature, just as is the use of the English language.

Psychologists tell us we can help develop our subconscious minds by mentally concentrating on one subject and physically engaging in another. For example, we might try reciting Lincoln's Gettysburg address while simultaneously writing the

Radiotelegraph CODE

Lord's Prayer — or better, while copying it on a typewriter. Practice copying code, on typewriter or in longhand, while carrying on a conversation with someone in the room. Send code with "bug" or straight key, while also engaged in conversation on another subject. This sort of mental exercise will help you separate your conscious thought from your subconscious reactions so that the latter may devote themselves to code copying.

Operating on the Air

- *General Operating Data*
- *Abbreviations*
- *Message Form*
- *Log-Keeping*
- *Radio Operator Licenses*
- *Call Signals*

THE radiotelegraph code is used for *record* communication. Aside from ability to copy at high speeds, then, a good operator is noted for his neatness and accuracy of copy. It is evident that an operator should copy what is sent, and if there is any doubt about a letter or word he has copied he should query the transmitting operator about it.

An operator with a clean-cut, slow, steady method of sending has a big advantage over the poor operator. Good sending is a matter of practice, but patience and judgment are just as important qualities of an operator as a good fist. Very often, transmission at moderate speeds moves traffic more quickly than faster but erratic sending. In hand operating, unusual words should be sent twice, the word repeated following transmission of "?". A transmitting operator who is notified of interference on his frequency, either static or man-made, should adjust his speed of sending to require the least number of "fills." Every transmitting operator should have facilities for monitoring to check the accuracy of his sending. Remember — accuracy comes *first*.

To this end, an operator copying in long-hand should use extreme care in writing, so there will be no chance of confusing an "l" with an "e," and the like. On a typewriter, the best practice is always to double-space between lines, write ten words to a line with an extra space or two after the fifth word in each line, triple-space between lines every fifth line. This is for the purpose of rapidly determining the number of words in a message as it is sent. As one gains mill-copying skill, he will be able to typewrite subconsciously in this pattern, an example of which is shown here as a portion of text:

IN FULL FORCE STOP THE ATTACK WILL BE SUPPORTED BY
BOMBARDMENT AVIATION WITH LIGHT AND MEDIUM TANKS IMMEDIATELY PRECEDING THE
ARTILLERY UNITS STOP ATTACHED TO EACH DIVISION WILL BE UNITS
OF THE SIGNAL CORPS FROM FORTMONMOUTH NEWJERSEY UNDER COMMAND OF
MAJOR J WORTHINGTON SMITH WHOSE DUTIES WILL BE SUPPLYING POINT
TO POINT COMMUNICATION FOR STAFF HEADQUARTERS STOP THIRTY FIVE HIGH
SPEED OPERATORS WILL BE REQUIRED BY EACH STAFF HEADQUARTERS FOR
THE VOLUME OF TRAFFIC EXPECTED DURING THESE

ABBREVIATIONS

To speed up radiotelegraph communication, a number of standard and special abbreviations have been devised. As time is a factor, uniform practices in operating are necessary to insure a ready understanding by both operators, so proficiency in the commonly used abbreviations is to be desired. Some of those prescribed by the regulations attached to the International Telecommunications Convention for all radio services follow:

C	Yes
N	No
W	Word(s)
AA	All after (used after a question mark to request a repetition)
AB	All before (similarly)
AR	End of transmission
AS	Waiting period
BN	All between
BQ	Reply to a request
CL	I am closing my station
CQ	General call to all stations
JM	Make a series of dashes if I may transmit
	Make a series of dots to stop my transmission
K	Invitation to transmit
MN	Minute(s)
NW	Now
OK	We agree (or It is correct)
RQ	Indication of a request
UA	Do you agree
WA	Word after (to be used after a question mark to request a repetition)
WB	Word before (similarly)
ADS	Address (similarly)
PBL	Preamble (similarly)
SIG	Signature (similarly)
TXT	Text (similarly)
SYS	Refer to your service telegram
CFM	Confirm (or I confirm)
MSG	Prefix to radiotelegram
REF	Reference to (or Refer to)
RPT	Repeat (or I repeat)
SVC	Prefix indicating a service telegram
TFC	Traffic
P	Prefix indicating a private radiotelegram
NIL	I have nothing to send you

XXX XXX XXX DE . . . urgent signal indicating message to follow regarding safety of mobile station or persons in sight therefrom (PAN is similarly used by aircraft);

TTT TTT TTT DE . . . , safety signal sent before meteorological warning messages and those concerning safety of navigation; SOS SOS SOS DE . . . , distress signal sent only by mobile stations in grave danger when requesting assistance (MAYDAY is the radiophone distress call similarly used).

There are also a number of abbreviations used by amateurs and, to some extent, by commercial operators. They originate with the old press codes in some cases; in others they have been concocted by individual operators and have found their way into universal use. We list on the following page some of the more frequently-encountered ones. While some are purely arbitrary, it will be noted that others are

LEARNING THE

simple phonetic spellings, some are the first and last letters of a word, some simply eliminate vowels, others use the letter X to replace part of a word, etc.:

ABT	About
AMP	Ampere
BCL	Broadcast listener
BK	Break
BN	Been, all between
BUG	Semiautomatic key
CK	Check
CL-CLD	Closing station; call; called
CUD	Could
CUL	See you later
CW	Continuous wave
DLD-DLVD	Delivered
DX	Distance
FB	Fine business, excellent
FREQ	Frequency
GA	Get answer, Go ahead
GB	Good-by
GBA	Give better address
GE	Good evening
GG	Going
GM	Good morning
GN	Gone, good night
GND	Ground
HI	Laughter, high
HR	Here, hear
HV	Have
HW	How
LID	"Lid," a poor operator
MG	Motor generator
MILS	Milliamperes
ND	Nothing doing
NIL	Nothing
NR	Number, near
OPN	Operation
OP-OPR	Operator
OT	Old-timer, old-top
PSE	Please
R	Are, all right, OK
RAC	Rectified alternating current
RCD	Received
RCVR	Receiver
RI	Radio Inspector
SED	Said
SEZ	Says
SIG-SG	Signature
SIGS	Signals
SINE	Sign, personal initials, signature
SKED	Schedule
TU-TNX	Thanks
TNG	Things
TMW	Tomorrow
UR-S	Your, you're, yours
VY	Very
WD-S	Word, words
WKD	Worked
WKG	Working
WL	Will
WAT	What
WUD	Would
WX	Weather
XMTR	Transmitter
YL	Young lady
73	Best regards
88	Love and kisses

Radiotelegraph CODE

In the text of a message, however, no words should be abbreviated by the operator unless they are so written by the sender. If the text includes punctuation, it should be spelled out in English.

MESSAGE FORM

Each radio communication service — commercial, amateur, military — prescribes its own message form, but basically all are generally similar to the example given below. An amateur message is broadly divided into four parts: (1) the preamble; (2) the address; (3) the text; (4) the signature. More specifically, a typical message contains the following elements:

- a) Number (of this message)
- b) Precedence (R, P or EMERGENCY)
- c) Handling instructions (optional)
- d) Station of origin
- e) Check (number of words in text)
- f) Place of origin
- g) Time filed
- h) Date
- i) Address
- j) Text
- k) Signature

The "precedence" indicates the relative importance of the message, as assigned by the originating station; usually it will be R, which stands for "routine." The handling instructions, which may or may not be included, give handling and delivering stations any special instructions. They use the letters HX followed by another letter, and sometimes a number. For example, HXA 10 means the message may be delivered by collect telephone if received by a station within ten miles of the addressee. A complete message sent by amateur radio c.w. might look like this:

NR 207 R HXB48 W1AW CK12 NEWINGTON CONN 2345Z OCT 25
HERBERT HOOVER JR W6ZH AA 890 SOUTH SAN RAFAEL AVE AA
PASADENA CALIF BT
BEST WISHES FOR A VERY HAPPY BIRTHDAY WITH MANY MORE
TO FOLLOW BT JOHN HUNTOON AR

This is the 207th message (usually they are numbered starting with number one at the beginning of the year and continuing until the end of the year, but the number is only a means of identifying the particular message), with a routine precedence (P for "priority" and EMERGENCY are used mainly in disaster communications). HXB48 means that the message should be cancelled if not delivered within 48 hours of the time it was filed. The station at which the message was originated is W1AW, the text of the message contains twelve words, and the place of origin is usually the location of the station of origin. The filing time is given in 24-hour Greenwich Mean Time. The date is given just as the month and day; occasionally you will hear some station give the year, but this is unnecessary and not usually included. Some stations will send "TO" before the address, but this also isn't really necessary. In the transmission of the address, the name of the addressee is separated from the street

LEARNING THE

address and the street address from the city of destination by the procedure signal AA (didahdidah) to avoid confusion. After the address and before the text you will hear the procedure signal BT (dahdidididah) to separate the address from the signature. Again, at the end of the text you will hear BT, and following the signature (some stations precede it by SIG, but this is not correct procedure) you will hear the signal AR (didahdidahdit), signifying the end of that message.

Quite frequently you will hear a station repeat a difficult word. This is usually indicated by the sending of the question mark (dididahdahdidit) after the word, then sending the word again. When the receiving station misses part of a message, he will indicate which part he missed by sending the last word received, a question mark, then the next word received, whereupon the sending station gives him the part between; or if the receiving station knows he has missed only one word, he may ask for WA (word after) or WB (word before) a certain word.

All amateurs may not always follow these ARRL standards, and in your listening on the amateur bands you may find variations of this procedure. Listening outside the amateur bands you will undoubtedly hear other procedures, although most commercial message handling is now done by radioteletype, facsimile and other methods not employing the International Morse Code. Your best bet to get code practice is to listen on the amateur c.w. bands; there you will get not only practice in copying, but you will develop tolerance for erratic sending habits of operators, interference from other stations and from atmospherics, fading, and familiarity with amateur customs and procedure which will make it easy for you to become one of the fraternity when you get the ticket.

LOG-KEEPING

FCC requires nearly every radiocommunication station to keep a complete operating record, or log, including such data as times and dates of transmissions, stations

DATE TIME	STATION CALLED	CALLED BY	FREQ ON DIAL	MY SIGNALS MET	MY SIGNALS MET	POWER WATTS	TIME USED HOURS	POWER WATTS	TIME OF RECEIVED GND	OTHER DATA
11-16-52										
6:15 PM	WOTQD	x	3.65	599	569X	3.5	A1	250	6:43	Tfc - rec'd 6, sent 10
7:20	CQ	x				7				
7:21	x	W4TW1	716	369	579				7:32	Vy heavy QRM on me
9:25	W9UKS	x	3.93	59	47	3.9	A3	100	10:05	Team J
11-18-52										
7:05 AM	MVKEHL	x	14.03			14	A1	250		Answered a W6
7:09	ZLZACV	x	14.07	339	559X				7:20	
7:21	x	KA2KW	14.07	469X	349				7:33	First KA
7:36	CQ	x								
7:37	x	W6TI	14.01	599	5890				8:12	

contacted, message traffic handled, input power to the transmitter, frequency used, and signature or "sine" of the operator in charge.

Log-keeping procedure differs with each class of communications service. A typical page from an amateur radio station log, prepared on the standard ARRL form, is shown above. Being that of the amateur service, the example here shown is quite free in style, yet it is illustrative of the form and data generally required.

Radiotelegraph CODE

RADIO OPERATOR LICENSES

Except in the military services, and with certain additional exceptions for some types of routine operations in railroad and aircraft radio, no one may operate a radio transmitter without possessing the necessary government operator license. The penalties for unlicensed operation are severe. Licenses are issued in the U.S. by the Federal Communications Commission, in two broad classes: amateur and commercial. As implied in the names, a commercial license permits its holder to operate a radio station for hire, while an amateur license is usable only for experimental communication "with a personal aim and without pecuniary interest." Under the Communications Act of 1934, FCC may issue licenses only to U.S. citizens. There are two recent exceptions. Aliens who hold aircraft pilot licenses valid in the U.S. may be issued licenses for aircraft radio stations. The U.S. has entered into reciprocal operating agreements with certain other countries, permitting the operation of amateur stations licensed by one of the parties in the territory of the other. For every license an examination is required, varying in scope according to the privileges conferred by the license. Licenses are normally issued for a term of five years, and may be renewed upon application if the applicant can show recent use of the license privileges.

An applicant for an amateur operator license must pass a code test of sending and receiving at the rate of five or thirteen words per minute, depending upon the class of license, and a written examination on radio rules and regulations as well as basic theory and practice. These tests are taken at one of numerous FCC examining centers, or by mail under the supervision of a licensed operator, again depending on the class involved. For more complete information on amateur licenses, you may write to the American Radio Relay League, Newington, Conn. 06111. Among other aids, the League publishes a companion booklet, *The Radio Amateur's License Manual*, which treats this subject in detail, including paraphrased questions and answers for the examination and the complete text of the FCC amateur regulations.

Commercial radio operator licenses are issued in several classes, with a different scope of authority for each. The First Class licenses carry the greatest privileges and naturally have the most inclusive requirements; at the lower end of the scale, in fact, the "licenses" are actually called "permits" or "authorizations" to indicate their lesser scope. FCC commercial written examinations are divided into elements, thus:

- Element 1—Basic law
- Element 2—Basic operating practice
- Element 3—Basic radiotelephone
- Element 4—Advanced radiotelephone
- Element 5—Radiotelegraph operating practice
- Element 6—Advanced radiotelegraph
- Element 7—Aircraft radiotelegraph
- Element 8—Ship radar techniques

Each license requires the passing of certain combinations of these elements, as follows:

Radiotelephone:

- Third Class—Elements 1, 2
- Second Class—Elements 1, 2, 3
- First Class—Elements 1, 2, 3, 4

Radiotelegraph:

Third Class — Elements 1, 2, 5; 16-w.p.m. coded groups, 20 w.p.m. plain language.

Second Class — Elements 1, 2, 5, 6; 16-w.p.m. coded groups, 20 w.p.m. plain language.

First Class — Elements 1, 2, 5, 6; 25-w.p.m. plain language, 20 w.p.m. coded groups; service endorsement on Second Class license.

In addition, there are several "restricted" classes of permits available in special cases for limited purposes.

By writing to the Federal Communications Commission in Washington, D. C., you may learn which is the local office nearest you. Write the latter for data on exact time and day of commercial examinations, and to make an appointment. Simultaneously request the necessary application blanks so you can arrange for notarization in advance. Each examination element requires between 1 and 1½ hours for completion, so allow yourself time accordingly. If you fail certain elements but pass others, you will automatically be issued a license of proper grade if the passed elements entitle you to it.

CANADA

In Canada, amateurs are licensed by the Department of Communications, Ottawa. A code speed of ten words per minute is required for the amateur certificate, fifteen for the advanced amateur certificate. There are also written and oral examinations. Details may be obtained from Ottawa or from the regional offices in Vancouver, Winnipeg, Toronto, Cité de Harve, P.Q., or Moncton, N.B.

CALL SIGNALS

Obviously every radio station needs a call signal to identify itself. In the case of military stations, these are generally arbitrarily-devised "tactical" calls which give no inkling of the location. The civil stations of the world, however, are assigned calls in accordance with an alphabetical table set up by the international telecommunications conferences, and thus it is possible to identify the nationality of a station by its call signal. "Blocks" of calls are made available to the various nations in accordance with prearranged tables. Thus, for example, any call beginning with W, K or N is seen to be a call of the United States. In many cases an initial letter is divided between two or more nations and it is therefore necessary to observe the first two letters before determining the nationality. For example, if the first two letters are XE, the station is Mexican, while if the first two letters are XZ, the station is Burmese.

In the United States, whenever a three-letter call is heard (sometimes followed by a number) it belongs to a land station. Ship stations have four-letter calls, although such calls are also frequently used for broadcasting stations. Aircraft stations have five-letter calls. The calls of amateur and experimental stations are made up of one or two letters, a single figure, and a group of not more than three letters.

Code Practice

- *Tone Sources for Practice*
- *A Code Instruction Table*
- *Practice Schedules*

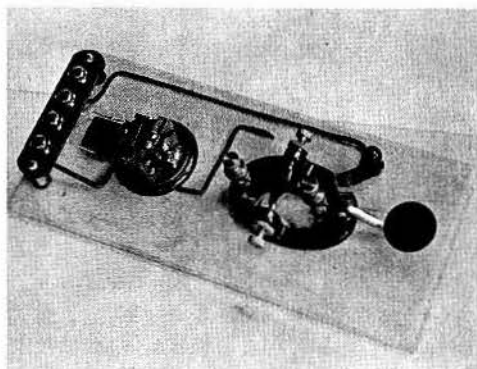
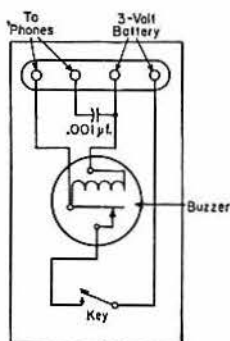
WHISTLING or "didahing" is only a substitute for the high-pitched sound of radiotelegraph code. Since a person beginning his code study without constant supervision is not capable of properly manipulating a telegraph key—should not touch it, in fact, until he thoroughly knows the code sounds—he can best learn by the spoken didah method. Once he progresses to the point where he is ready to learn how to send, he needs a tone source which can be keyed for sending practice. Perhaps the best way is to have two people learn the code together and send to each other by means of a buzzer-and-key outfit. An advantage of this system is that it also develops sending ability, for the person doing the receiving will be quick to criticize uneven or indistinct sending. If possible it is a good idea to obtain the assistance of an experienced operator, so that you will learn how well-sent characters should sound.

Components of simple oscillators may be obtained from local radio stores, or perhaps from the "junk box" of a radio experimenter friend.

STONE SOURCES FOR PRACTICE

The simplest possible code-practice set consists of a key, buzzer and dry-cell battery. The buzzer should be one generating a high-pitched tone, such as the E. F. Johnson type 114-400. The assembly shown in the accompanying photograph and

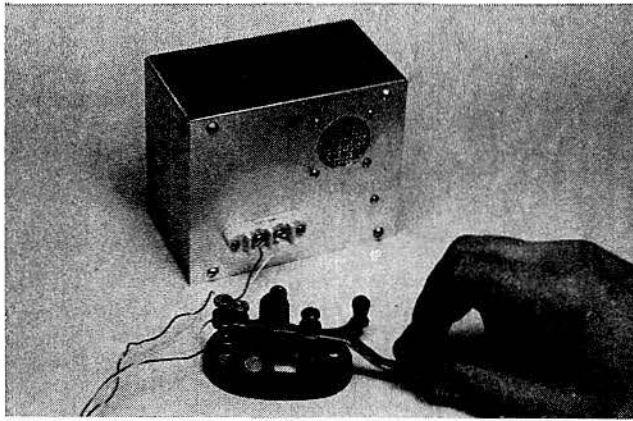
Photograph (right) and circuit arrangement (left) of a buzzer code-practice set. The headphones are connected across the coils of the buzzer with a fixed capacitor in series. The value of this capacitor determines the strength of the signal in the headphones. If the value shown gives an excessively loud signal, it may be reduced to 500 pf. or 250 pf.



LEARNING THE

diagram is slightly more elaborate, in that it includes a capacitor and pair of terminals for headphones. Headphones give a little more realism to code practice and help to shut out extraneous sounds that may be distracting. The headphone connection and capacitor may be omitted, however, if they are not needed. The battery can consist of two No. 6 (or other reasonably heavy-duty) dry cells connected in series. Alternatively, a 3-volt lantern battery can be used.

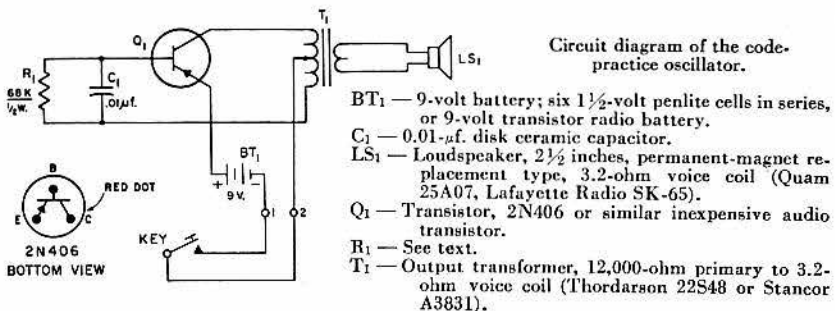
A better tone, more nearly identical to the audio beat note a radio operator reads, is that produced by an audio oscillator. The battery-operated transistor audio oscillator shown on these pages produces enough volume for code practice in small groups and is easy to construct. It consists of a transistor, a capacitor, a resistor, an output transformer and speaker, and a dry-cell battery, and is built on one side of a



Transistor audio oscillator ready for code practice. The small speaker puts out enough volume for group practice.

3 × 4 × 5-inch aluminum box. A four-terminal tie point is used for mounting C_1 , R_1 , Q_1 , and for connecting the leads from T_1 . Special care must be taken when soldering the transistor leads, as too much heat can ruin the transistor. Use a pair of long-nose pliers to hold the lead being soldered, grasping the lead close to the transistor body. The pliers will absorb most of the heat before it can reach the transistor.

Practically any inexpensive transistor will be suitable for this circuit. The one shown in the photograph is a 2N406. If this type is used, the lead closest to the red dot (collector) on the body should be connected to one end (either one) of the



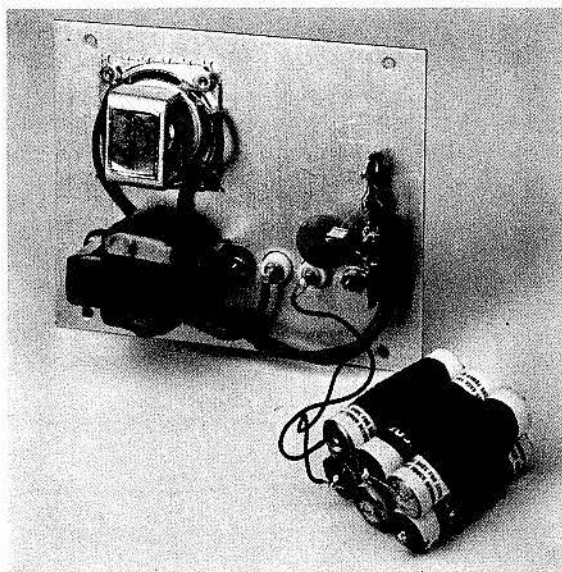
Radiotelegraph CODE

primary winding of T_1 . The other end of the primary winding goes to C_1 and R_1 as shown in the circuit. The center transistor lead (base) should be connected to the junction of C_1 and R_1 . The remaining lead (emitter) goes to the positive end of the battery.

The pitch of the audio tone can be lowered by changing R_1 to 47,000 ohms, or raised by using 100,000 ohms at R_1 .

The construction shown in the photographs does not have to be followed exactly. Also, a larger speaker can be substituted; the 3½-inch size, which is available at practically any radio parts store, will work equally well. If a different type of transistor is used, make sure that it is one having PNP construction and that the leads are connected as described above.

The speaker is mounted directly above the output transformer T_1 . At the right of T_1 is the two-terminal strip (Millen 37302) for the key connections. The remaining components are mounted on the four-terminal tie point.



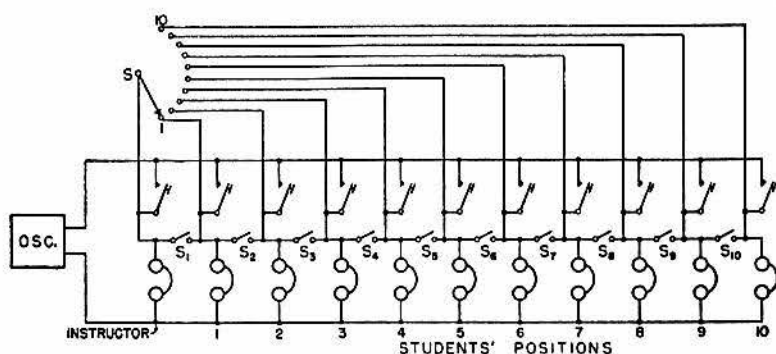
A CODE-INSTRUCTION TABLE

The preceding material has been written primarily for the individual who must learn his code with little or no outside help. If a number of persons wish to learn the code as a group or develop speed after learning, and if a *competent* instructor can be located, the best method of instruction is by means of a code table (page 34).

Any such table should be so wired as to permit the instructor to send to the whole class, but by a little special wiring many other things are possible. In the one shown here, for instance, each student can practice sending, independently and to himself; yet the instructor, by means of the selector switch, can listen in on each student in turn, can break in on him and correct his errors. Moreover, simply by throwing a few switches, facing pairs of students may join in independent two-way conversations, and each such pair may be monitored in turn by the instructor—all on the one table and from the common oscillator. Bigger "nets" can be switched at will, while other students at the same table continue practice either singly or in pairs, all under control and supervision.

The instructor should sit at one end of the table, facing down its length, the students arranged along each side. A center partition and cross-partitions make a sort of "private office" of each operating position, each with its phones, key and switch.

LEARNING THE

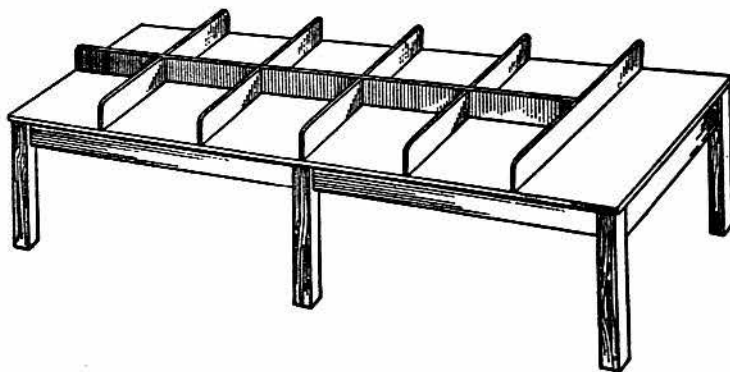
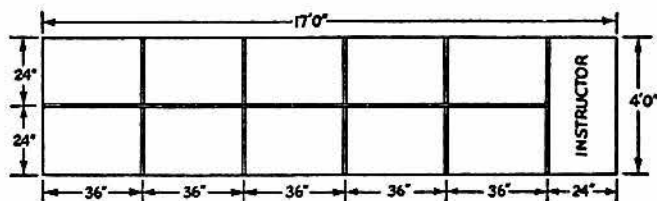


Wiring diagram of flexible multiposition code-instruction table.

The oscillator is at, or handy to, the instructor's end of the table, his controls on a low panel in front of him, running across the table.

At each student's position, a key and headset in series are bridged across the line carrying the continuously-running audio tone. The headsets (and keys) may be paralleled by closing the interconnecting switches, each switch being associated with the position of the same number. A tap from each position is taken off to the multipoint switch (S) at the instructor's position so that the latter can also place his headset and key in parallel with those at any of the ten positions.

When the instructor sends to the entire class all the small switches are closed, connecting all positions in parallel. With all switches open each student can practice



A suggested form of construction for the code-instruction table. Dimensions may be varied to suit individual needs. Legs may be made of 4×4 -inch stock; top and side rails of 1-inch boards. Appropriate bracing should be provided underneath.

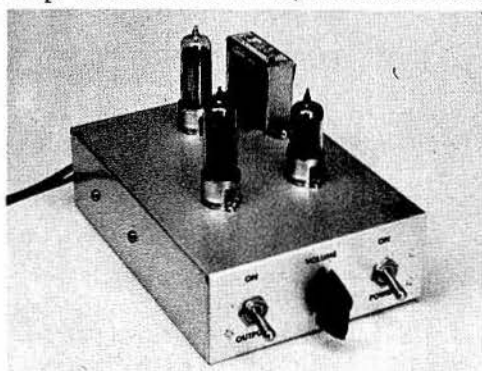
Radiotelegraph CODE

sending by himself, and the instructor can listen to any student by setting the selector switch, S_1 , to the appropriate position. This also puts the instructor in parallel with the student at that position so that the instructor can "work" the student.

Suppose now that students are to practice together in groups of two. Closing S_2 connects Positions 1 and 2; closing S_4 connects 3 and 4; S_6 connects 5 and 6, and so on. The odd-numbered switches would be open in this case. The instructor can listen in on any group by setting the selector switch to either position in the group, and again can break in on the work. Larger groups can easily be formed; for instance, closing switches Nos. 2, 3 and 4 will connect Positions 1, 2, 3 and 4 together; closing S_6 and S_7 will connect Positions 5, 6 and 7 together, and so on.

The number of positions can be extended indefinitely by following the same wiring system. Separate tables, grouping perhaps ten students to a table, can be used if the

TABLE I		
Freq. (c.p.s.)	C_1, C_2, C_3 (pf.)	R_1, R_2, R_3 (ohms)
400	400	.68 meg.
600	400	.33 meg.
800	400	.22 meg.
1000	400	.39 meg.



The code-practice oscillator for instructing large groups is built on a 2 × 5 × 7-inch aluminum chassis. All wiring is insulated from the chassis. The volume control is centered between S_1 and S_2 . L_1 , the filter choke, is located to the rear of the 35W4.

room is too small to accommodate a single table for a large class. In such a case the instructor could have a central position with a selector switch for each table.

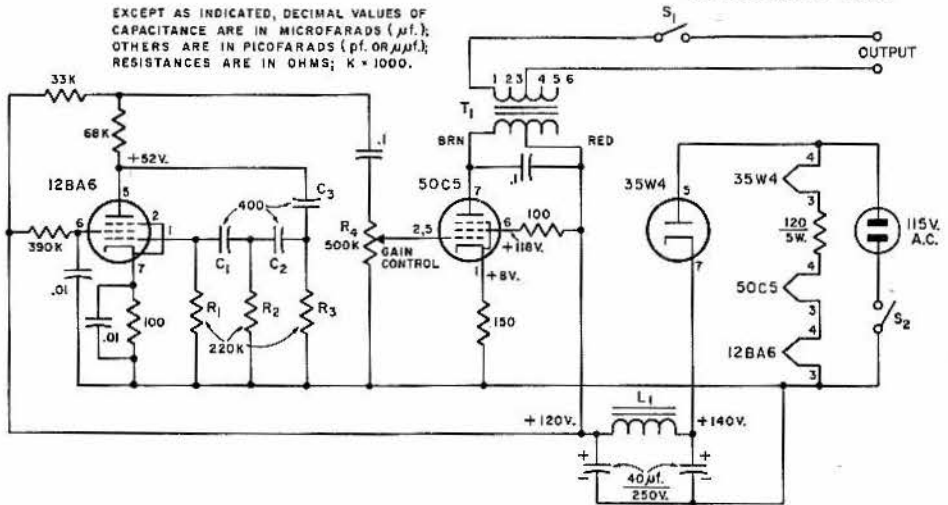
The interconnecting switches, S_1, S_2 , etc., may be ten-cent-store s.p.s.t. knife switches and the instructor's selector switch a wafer-type unit. The latter can be obtained in the single-pole type with as many as 23 contacts. Since no special precautions need be taken with respect to insulation or voltage drop, probably the most economical wiring job can be done by using ordinary bell wire. In cases where the students bring their own headsets it would be advisable to mount a tip-jack assembly at each position to facilitate connection.

The code-practice oscillator described earlier in this chapter is suitable only for two or three sets of 'phones; it is, therefore, necessary to build a special type of oscillator for use with the code table. A circuit diagram suitable for handling up to thirty or forty headsets is shown on the following page. Operating directly from the 115-volt line, it consists of a 12BA6 oscillator followed by a 50C5 power amplifier and 35W4 rectifier.

The pitch of the tone is determined by the constants of the RC circuit, $R_1R_2R_3-C_1C_2C_3$. A tone of approximately 800 cycles is obtained with the values shown in the schematic of the unit. Table I lists values of capacitance and resistance that should be used if a tone in the 400- to 1000-cycle range is desired.

When wiring the output transformer, T_1 , into the 50C5 circuit, employ taps

LEARNING THE



Circuit diagram of a code-practice oscillator suitable for instructing large groups. Except as indicated, fixed resistors are $\frac{1}{2}$ watt. Capacitors with polarity shown are electrolytic; 0.1- μ f. capacitors are tubular paper; 0.01- μ f. capacitors are disk ceramic; others may be either ceramic or mica.

C_1, C_2, C_3 — See text.

L_1 — Universal-type filter choke; approximate ratings (not critical), 4 to 10 henrys, 50 ma.

R_1, R_2, R_3 — See text.

R_4 — 0.5 megohm control, audio taper.

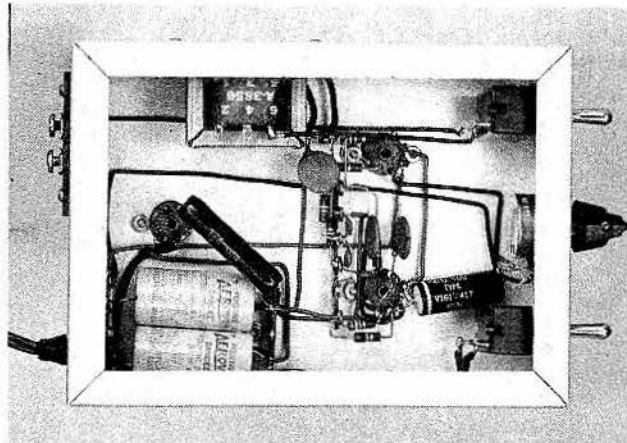
S_1, S_2 — S.p.s.t. toggle.

T_1 — Universal-type adjustable-ratio 4-watt output transformer (Stancor A-3856 or equivalent). One-half of primary winding used. Secondary connected for approximately 2 ohms.

that result in an approximate primary-to-secondary impedance ratio of 2,500 to 2 ohms. This impedance ratio results in adequate audio output for 30 or more sets of low-impedance 'phones and prevents noticeable fluctuations in audio volume as the number of active headsets is varied over a wide range.

For eliminating danger of electrical shock, the complete code-practice circuit must be isolated from the metal chassis. The voltages to be expected at various points in the circuit are shown on the schematic. The negative lead of the voltmeter must be connected to the common negative bus (the negative side of one of the 40- μ f. filter capacitors is a good place) rather than to the chassis during the voltage measurements.

A bottom view of 12BA6-50C5-35W4 code-practice oscillator. The output transformer is mounted on the side wall of the chassis. Output terminals and a grommet for the line cord are located at the rear of the unit.



Radiotelegraph CODE

The output switch, S_1 , enables the instructor to cut off the tone from the entire class, and thus serves as a simple means of attracting attention when group work or individual sending practice is being carried on.

AMATEUR CODE PRACTICE

To assist beginners to obtain amateur licenses and amateurs in upgrading themselves, and to encourage better code proficiency for all operators, the American Radio Relay League conducts nightly code-practice transmissions from its headquarters station W1AW in Newington, Conn. Practice is transmitted three times daily Monday through Friday, twice daily on Saturday and Sunday. The first practice transmission is at 9:00 A.M. EST/EDST (1400/1300 GMT); on Monday, Wednesday and Friday speeds are 5, 7½, 10, 13, 20 and 25 wpm; on Tuesday and Thursdays speeds are 35, 30, 25, 20 and 15 wpm, both in the order stated. The second transmission is at 7:30 P.M. EST/EDST (0030 GMT the following day or 2330 GMT the same day ¹) at speeds of 10, 13 and 15 wpm *every* day. The third transmission is at 9:30 P.M. EST/EDST (0230/0130 GMT the following day ¹); on Monday, Wednesday and Friday, speeds are 35, 30, 25, 20 and 15 wpm; on Sunday, Tuesday, Thursday and Saturday, speeds are 5, 7½, 10, 13, 20 and 25 wpm. The transmissions are simultaneous on 1805, 3580, 7080, 14080, 21,080, 28,080, 50,080 and 145,588 kHz.

COMMERCIAL PRESS SCHEDULES FOR PRACTICE

Students of code wishing to increase their copying abilities can make valuable use of transmissions by commercial and military stations at various speeds. A list of a few good sources of practice is here given. Students making use of these and other transmissions are reminded that *addressed information may not be divulged except to the addressee*, and the Communications Act provides severe penalties for violators. A more complete list is available from ARRL Hq. Send self-addressed stamped envelope. Do not use such transmissions for anything but code practice. Since these schedules are subject to change, please consult *QST* for reliable up-to-date information.

Time (GMT)	Call	Origin	Frequency (kHz)	Speed	Schedule
0300	WSL	New York	4343, 6418	25	Daily
1335	WAX	Florida	8526	16-25	Daily
1605	WOE	Florida	8486	16-25	Daily
0130	NAV	Wash., D. C.	2025, 4015 7301, 14385	20	Sun.
0130	WAR	Wash., D. C.	3347, 5217	15-20	Tue.
0430			6997.5, 14405 20994	25	
1800	AIR	Wash., D. C.	4873.5, 7441.5	15	Wed. &
0200			13995		Thur.
1800	AG4FL	Kansas	7305.5	15	Wed.
0200	AG4FL	Kansas	4591.5	15	Thur.

¹ Much amateur work is conducted in terms of Greenwich Mean Time (GMT). This is a 24-hour clock system in universal use, in which 0000 (or 2400) represents midnight and 1200 is noon. Persons in the Central Standard time zone are six hours behind GMT. For example, 0100 GMT Tuesday would be 1900 (7:00 P.M.) CST Monday. EST is five hours behind GMT, MST seven, PST eight. "Daylight saving" time is one hour faster in each case.

Class Instruction

• Exercises in Receiving and Sending

THE material in this book has been prepared as a text for the individual student, principally for home-study purposes. This appended section is to explain briefly the classroom technique necessary for code instruction and to present a suggested set of lessons that a class may follow.

The classroom student should be taught in terms of sound at all times. While he should understand the use of the terms dit and dah, they need not be used in learning letter sounds; the actual code sound of the letter in his headphones should be the student's first conception. It is obvious that only a reasonably-proficient operator should be allowed to teach; students will acquire any improper habits of a poor operator/instructor—which, once learned, will prove extremely resistant to change.

A class in code should have the equivalent of the code-training table described in the chapter on "Code Practice," with one operating position for each student up to a total of 20, the maximum which can be taught *efficiently* by one instructor. Later in the course a code tape machine, or a communications receiver to tune in slow-speed transmissions, will be found quite helpful since while one is in use the instructor may make the rounds of the class, offering individual help.

The first code sounds should be formed at a speed of approximately 15 words per minute, with reasonably large spaces between letters to make the over-all total about 5 words per minute. By forming characters at a rapid rate, they are more likely to be learned as complete sounds instead of combinations of dits and dahs. As the class progresses, spaces between words can be shortened to increase the over-all speed. The speed of transmission may be estimated by counting the total number of letters sent in one minute and dividing by 5.

No certain number of hours has been specified for the course. Instructors may take up a new lesson when approximately four-fifths of the class can copy or transmit perfectly 50 consecutive letters of the lesson under study. Some lessons may be completed in an hour's class time; others may require five or six hours.

LESSONS IN RECEIVING

Ten lesson groups follow, each containing adequate practice material for classroom use. It will be noted that the order of learning letter sounds is somewhat different here than in the opening chapter. Organized class training under a competent instructor avoids certain home-study difficulties which are best met by presentation of letter sounds beginning with 5 and 0. The classroom system presents scientifically to the new man an interesting approach to learning the code: by taking up most-used letters first, many words and even sentences can be utilized right from the first lesson, so the student can see his progress from the start. The value of code groups has not been neglected, however; the last exercise in each lesson presents adequate coded material. The student's progress can best be graded by means of these code groups.

In each lesson the procedure should be: (1) review the work of the previous lesson; (2) send the new characters slowly in no particular order, to accustom the students to the sound of each; at first, name the English character while its code-sound is being sent; (3) send the characters of the previous lesson indiscriminately mixed with the new ones, slowly and in no particular order; (4) send the characters of both lessons in the form of words and sentences, gradually increasing their lengths.

LESSON ONE . . .

The instructor should spend a preliminary period lecturing on the introductory material in the first chapter of this book, including the timing and rhythm exercises, but using the tone source along with *didah* and whistling.

LESSON TWO . . .

Characters used: E T O A N I S

Practice material:

AN	NO	ON	IT	TO	SO
IS	OAT	TAN	TEN	ONE	ANT
NIT	TOT	NET	TIE	ETA	TIN
TOE	SON	SIN	SIT	SET	ASS
SEA	INTO	TENT	STONE	ANON	NINE
NEAT	TOTE	TOOT	NONE	TEAT	TINT
ETON	SEAT	EASE	SENT	TAINT	STINT
EATEN	STONE	TOAST	TEASE	SNOOT	SENSE

- | | |
|---------------------|-----------------------|
| a) SIN NO ONE | f) IT IS NOT TEA |
| b) IT SENSES TASTE | g) ETA IS A NOTE |
| c) NONE IS SENT | h) NO SENSE IN EASE |
| d) SNOOTS EATS OATS | i) STAINS TINT A SEA |
| e) NO SEASON IS SET | j) ASSES NET NO SENSE |
- k) SONNIE SAT ON A TIN SEAT
 l) IT TOASTS IN TEN TASTES
 m) AN OASIS IS NEAT EASE
 n) TIE A NINETEEN TON STONE TO A TENT
 o) TOAST IS EATEN AT ETON
 p) STAINS TAINT ONES NEATNESS
 q) TOTE A STONE TO TENNESSEE

ETO	ANI	SIN	AOT	ESI
SAOT	NITN	SOOE	IATE	ANOE
SINES	TOEIS	NAOTE	ONTIS	ESTAO
ENTOIS	TOOSEI	OENTST	ANSENO	NANOSE

LESSON THREE . . .

New characters used: R H D U C M L

Practice material:

RAT	RUN	RUT	RICE	RUSH	RASH	RADIO
-----	-----	-----	------	------	------	-------

LEARNING THE

HAD	HUM	HAM	HAND	HELM	HULL	HOUSE
URN	USE	NUT	UHIT	UNDO	URAL	UNDER
DAD	DAM	DON	DUDE	DICE	DOLT	DUNCE
COT	COD	CAM	CHIT	CORD	CURT	CHORD
MAN	MAD	MET	MOSS	MICE	MODE	MODEL
LAD	LET	LID	LEAD	LORE	LUST	LURID

REDUCE	RADIUS	RANCID	RADICAL	MARCONI
HALTER	HUDDLE	HORROR	HUMIDOR	ULTIMATE
ULSTER	UNLACE	URCHIN	UNARMED	STATION
DOLLAR	DISCUS	DANCER	DUCTILE	RESISTOR
COLLAR	CRUISE	CUDDLE	COLLIER	ANTENNA
MUDDLE	MURMUR	MOTHER	MIRACLE	AERIAL
LEADER	LANCER	LUSTRE	LECTURE	COIL

TRANSMITTER	CHROMOSOME	MEDITERRANEAN
RADIATION	DASTARDLINESS	NECESSITARIANISM
OCCLUDED	CHROMIUM	OMNISCIENCE
ANTHRACITE	EDUCATIONAL	RESIDENTIAL
AMERICAN	THERMITE	SANITARIUM

- a) HAMS ARE ACTORS TOO.
- b) MIRACLES SELDOM OCCUR.
- c) MUSIC HATH LOTS TO CHARM.
- d) MAD CATS MURDER MICE.
- e) SEND CODE AND SUCCEED.
- f) SOME DANCERS ARE CRUDE.
- g) HENS HATCH SMALL ROOSTERS.
- h) MORSE CREATED THIS MIRACLE.
- i) CURSES ON ALL HEARSEs.
- j) RURAL HOUSES ARE HUMID.
- k) ANTHRACITE COAL IS HARD.
- l) RESISTANCE IS MEASURED IN OHMS.
- m) UNCLE SAM NEEDS US ALL.
- n) THE SUN SHINES IN CONNECTICUT.
- o) CODE CLASSES ARE LITTLE TOIL.
- p) CHARACTERS ARE DOTS AND DASHES.
- q) CODE IS HEARD NOT SEEN.
- r) MARCONI LECTURED ON CURRENT RADIATION.
- s) INTO DEATH MARCHED THE THREE HUNDRED.
- t) AMATEUR RADIO IS AS OLD AS THE RADIO ART.

URT	HLU	ADC	CIM	SLE
LTRH	OAUD	NICM	SRHL	MAUE
HSDNM	RHUAC	ODNCL	SICDO	IUHER
TCODME	ADHRHU	CNDUTR	SLMIRU	LSEIRC

Radiotelegraph CODE

LESSON FOUR . . .

New characters used: P F W Y G B

Practice material:

FOW	PEP	PIG	PAGE	PYRE	PITY	POPPY	PUFFY
POP	FRY	FIG	FROG	FLOW	FILE	FABLE	FOGGY
WEB	WON	WHY	WHIG	WOLF	WING	WHIFF	WEIGH
YIP	YOU	YEG	YOWL	YELP	YAWN	YOUNG	YACHT
GOB	GYP	GIN	GLOW	GASP	GRUB	GABLE	GLOBE
BOY	BEG	BIT	BANG	BEEF	BLOB	BADGE	BEFOG

PACIFY	PEBBLE	PHLEGM	PEPPERY	AFFLICT
FEEBLE	FIDGET	FLINTY	FAIRWAY	BABYLON
WAFFLE	WEAPON	WINDOW	WAYWARD	CABBAGE

DEPOSITORY	ELECTROTYPE	FISHWIFE
PYGMY	GUYWIRE	TAPE
ALPHABET	PRACTICE	MESSAGE
SUPERHET	GEAR	SIGNAL
CAPACITY	FLYWHEEL	FILTER
CRYSTAL	PURPOSE	STABILITY
FACTOR	BEATNOTE	AMPLIFIER

- EPIGRAMS SAY MUCH IN FEW WORDS.
- WHAT IS WORTH DOING IS WORTH DOING WELL.
- FORTUNE BEFRIENDS THE BOLD.
- ILL BLOWS THE WIND WHICH PROFITS NOBODY.
- WINGS FLY FAST.
- WHIFFS OF FOG PACIFY FEEBLE AFFLICTIONS.
- BOYS YELP WHEN BIT BY DOGS.
- A PUFFY BLIMP IS A PEPPERY FISHWIFE.
- TETRODE AMPLIFIERS RELAY POWERFUL SIGNALS.
- RUBBER WHEELS AFFECT RIDING COMFORT.
- PUT YOUR FLYWHEEL IN GEAR.
- GASP AND YOWL BUT BUY DEFENSE BONDS.
- WHOOOP IT UP BEFORE DYING.
- LIGHT GLOBES GLOW WITH POWER.
- TIGHTEN CAP BEFORE TIPPING.
- THE BADGE OF COURAGE IS WON BY PLUGGING.
- ALWAYS COPY BEHIND IF POSSIBLE.
- BICYCLE RIDING CAUSES PUFFING.
- COPYING BY TYPEWRITER IS GOOD PRACTICE.
- PYGMY GUYWIRES PUT STRENGTH INTO POLES.

HFM
FHEP

ODW
PRTF

AUY
YUOW

NCG
WDAY

IMB
BMNG

LEARNING THE

BCADF
PSDGCB

GUOHP
FUIYMW

YDTRB
AWPFLY

WHEMG
GHORBG

FRSCY
WRNFY

LESSON FIVE . . .

New characters used: J K Q X Z V

Practice material:

JET	JUNK	JAVA	JERK	JACOB	JAPAN	JELLY
ASK	KINK	KEEP	KALE	KHAKI	KNACK	KNAVE
QUO	QUIT	AQUA	QUIZ	QUICK	QUAIL	QUEEN
JIM	OXEN	AXLE	XRAY	EXACT	AXIOM	OXIDE
ZOO	ZERO	JAZZ	ZING	AZURE	CRAZY	ZEBRA
IVY	VINE	HAVE	VOID	VIVID	AVAIL	EVERY

JACKAL	JERKIN	JOVIAL	JONQUIL	ZENITH
KIBITZ	KINGLY	KAISER	KICKOFF	EMBEZZLE
QUARTZ	QUAVER	QUORUM	ACQUIRE	PULVERIZE
PICKAX	EXCUSE	XERXES	EXPLAIN	GRAZE
ZEPHYR	ZIGZAG	ZOUNDS	BUZZARD	OXYGEN

- a) JACOB ASKED THE KING FOR QUARTZ OXIDE.
- b) QUICK AXIOMS AVOID UNJUST TAXES.
- c) QUIZZES QUICKEN THE REFLEXES.
- d) JOVIAL KIBITZERS ARE VERY OBNOXIOUS.
- e) JAPANS JUNK QUILTS QUICKLY.
- f) EVERY QUEEN KEEPS JONQUILS.
- g) HEXAGONAL KNUCKLES HAVE CRAZY KINKS.
- h) XERXES EXPLAINED THE QUICK VICTORY.
- i) VELVET QUAIL GRAZE IN ZOOS.
- j) KINGLY BUZZARDS QUAVER IN AZURE SKIES.

JPR	SNQV	QYLGV	ZKJFBG
FKH	JAHK	XWOFZ	RHKPVO
DQW	ZGVM	KPCAJ	QLFPDQ
XYU	YXUT	BZLNK	XMZNUJ
ZGM	ZWDO	IVKQW	YKWZBG
BVC	KBLI	YJCTV	HSIXJA
QPX	JFRE	EQSXR	MCVXZQ

Beginning with the next lesson, instructors may make good use of newspaper and other text for practice material.

LESSON SIX . . .

New characters used: 1 2 3 4 5

Practice material:

215	3242	34125	244521
-----	------	-------	--------

Radiotelegraph CODE

431	2151	42153	352142
234	4531	33214	332312
551	5324	24115	545134
422	1543	53425	152342
342	3154	14523	453152
351	2231	21435	514324
AE1	CP2X	3Z4BJ	123VJQ
3T5	14B3	XYZ51	W3X5AM
420	F3G4	3KP14	435PT1
N43	EAR5	RY311	AW1JOZ
2MC	424V	14V4U	2UV4JB
HT5	D1J3	L3VJ4	XU4PY1
425	412V	21JJ2	5SH32W

LESSON SEVEN . . .

New characters used: 6 7 8 9 0

Practice material:

867	9768	77689	967760		
906	0069	90870	806970		
760	7987	68096	688979		
896	0869	86970	087068		
707	8776	09906	786096		
998	8609	87780	779680		
807	7068	87669	696087		
196	3874	62840	647359		
245	1928	17395	821073		
837	5603	61723	489625		
604	7495	89450	107446		
932	1620	42417	829310		
758	4835	38950	593758		
103	2071	59636	621054		
C3B7	Q78G	F3M0J	Y9UBC	A1B3C2	9105P4
JW19	H5SH	132R5	WJV41	4D6F5E	Q6U3V4
47BE	4CY3	9L4PV	0OX23	9HC7JI	SM3801
X6Z2	ZK67	K176B	PQ954	8L0KM3	607GT

LESSON EIGHT . . .

At the conclusion of the seventh lesson, most students should be able to recognize letters and numerals immediately when heard. If not, more practice is indicated, especially on the characters causing trouble. It will be helpful to send some letters frequently confusing to the beginner such as G-W, F-L, O-S, P-X, Y-Q.

GOWN FILM ARK JAW OASIS APEX QUERY
WAGER LEAFLET WORKER BLOSSOM EXPERT

LEARNING THE

The following sentences employ every letter of the alphabet:

- a) THE EXPLORER WAS FROZEN IN HIS BIG KAYAK JUST AFTER MAKING QUEER DISCOVERIES.
- b) WHENEVER THE BIG BLACK FOX JUMPED THE SQUIRREL GAZED VERY SUSPICIOUSLY.
- c) MY HELP SQUEEZED IN AND JOINED THE WEAVERS AGAIN BEFORE SIX O'CLOCK.
- d) WE DISLIKE TO EXCHANGE JOB LOTS OF SIZES VARYING FROM A QUARTER UP.
- e) A QUART JAR OF OIL MIXED WITH GOOD ZINC OXIDE MAKES A VERY BRIGHT PAINT.
- f) WHILE MAKING DEEP EXCAVATIONS WE FOUND SOME QUAIN T BRONZE JEWELRY.
- g) SIX JAVELINS THROWN BY THE QUICK SAVAGES WHIZZED FORTY PACES BEYOND THE MARK.
- h) THE PUBLIC WAS AMAZED TO VIEW THE QUICKNESS AND DEXTERITY OF THE JUGGLER.
- i) WE QUICKLY SEIZED THE BLACK AXLE AND JUST SAVED IT FROM GOING PAST HIM.
- j) THE JOB REQUIRES EXTRA PLUCK AND ZEAL FROM EVERY YOUNG WAGE EARNER.

LESSON NINE . . .

New characters used: Comma, period, interrogation, double dash ($\overline{BT}$), end of message ($\overline{AR}$), end of communication ($\overline{VA}$), wait ($\overline{AS}$), fraction bar ($\overline{DN}$).

Practice material:

Continue practice of plain text, now including punctuation. It is not necessary to conduct instruction in more rare punctuation until the student progresses somewhat further. Also, send the students some messages in standard form, using all procedure signals necessary in the process (see chapter on operating). Send some with texts in the form of cipher groups as well.

LESSON TEN . . .

To acquaint the student with receiving conditions as he will find them in actual radio communication, play a couple of broadcast receivers while giving the students practice, or have some of the class engage in audible conversation.

If communications receivers are available, tune in code signals that can be heard while practice is going on. Decrease volume of the practice-tone source. Vary tone and stability of the source. Use commercial press schedule transmissions for code practice.

Radiotelegraph CODE

LESSONS IN SENDING

The chapter on "Learning to Send" should be thoroughly reread at this point. It will serve as an outline of lesson material, additional practice exercises being given herewith. When taking up new material, the instructor should first send the characters he wishes his students to learn, and then have them repeat individually after him. Characters can then be combined to form words, sentences and code groups for additional practice. Here again, the instructor should thoroughly review a preceding lesson before attempting new material. While learning to send, the student's speed of transmission should be kept low until he demonstrates ability to send all the characters correctly.

LESSON ONE . . .

This period should be devoted to learning proper posture, grip of key, hand and wrist movements, and exercises in sending multiple dits and dahs, as outlined in the sending chapter.

LESSON TWO . . .

Characters used: E I S H 5 T M O Ø

Exercises:

SEE	MOST	SHEET	MISSES	HEMOTH
HOT	HISS	THEME	THESIS	HOSTESS
SIT	EMIT	MOIST	ESTEEM	SOMETIMES
HOE	TOSS	HOSTE	SMOOTH	MOTTOES
OHM	THEM	SHOOT	SETTEE	SIMMER
ØØ5	Ø55Ø	5Ø5Ø5	55ØØ5Ø	5Ø5Ø5Ø

- a) SEE ME HOME SOME TIME.
- b) HIS TEETH SEEM A MESS.
- c) MISS THIS SHOT.
- d) THIS IS HIS HOME TOO.
- e) I SEEM TO OMIT IT SOMETIMES.
- f) OTTO MEET HIM THIS TIME.
- g) HE OMITTS THE ITEMS HE MISSES.
- h) THE HOE HIT ME THIS TIME.
- i) THIS IS OTIS SHOE.
- j) HE HIT HIS HOSTESS 5Ø TIMES.

LESSON THREE . . .

New characters used: A R L W J I P

Exercises:

LIL	WILI	LAIR	PLATE	HAWSER	PALATE
OWL	PASS	LAME	SPELL	ATTIRE	RAPIER

LEARNING THE

LAW	MAIL	PAIL	ALERT	HAWAII	WALLOP
APT	REEL	PEAR	JEWEL	ASHORE	RESISTOR
PAR	TRIP	RAIL	REPEL	ARTIST	LASSOES
JAP	LAST	WASH	RALPH	JESTER	JEEPERS
501	JOWL	TRIAL	RATHER	JEWISH	JARMINE
JIM	5105	50101	515010	PALLOR	05100

- RALPH PASS THE PLATE.
- ALERT HAWAII RESISTS JAPS.
- THE LAW REPELS LAME JESTERS.
- HE WILL LET LIL SPELL RESISTOR.
- REAL AERIALS ARE JEWELS.
- JEWISH HATS ARE APT ATTIRE.
- RATHER PALE PALLOR APPEARS AT HIS JOWL.
- HAWSER REELS WASH ASHORE.
- PHILIP WALLOPS THE ARTISTS JAW.

LESSON FOUR . . .

New characters used: U F 2 V 3 4

Exercises:

AVE	FIVE	HUMUS	MUSEUM	SITUATE
EVE	FUSE	STOVE	FERVOR	ISTHMUS
SUE	FIFE	SHAFT	RAFFLE	FAVOR
IOU	VILLE	UNFIT	FAMOUS	FEVER
OFF	JAVA	SIEVE	VELLUM	SHOUT
FLU	SURE	RUFUS	VERSUS	FRUIT
VIA	OVUM	PUFFS	FURROW	VOLUME
234	4132	50312	154302	123450

- FAMOUS FRUITS PUT FLAVOR INTO LIFE.
- SUE SURE HAS FLUE.
- EVE MUST VOTE VIA RUFUS.
- ASSUME THE STATUS OF SOVIET RUSSIA.
- AUTOISTS MOUNT VAST SUMMITS.
- LET US VETO THE ISSUE OF UTAH.
- FROM HUMUS ISSUES SCUM AND MUSS.
- MUFFLE THE SHOUTS FROM JAVA.
- PURSUE THAT AUTO SOUTH.
- OUR SUITS ARE SURE FAMOUS.

LESSON FIVE . . .

New characters used: N D B 6 8 9 X

Exercises:

BOX	698	OXEN	OXIDE	10964	DEXTER
-----	-----	------	-------	-------	--------

Radiotelegraph CODE

NED	BOND	6289	BANAL	BUBBLE	EXPAND
DEN	NEXT	DUMB	DEBUT	DIDDLE	401523
BUS	DRAB	DIXIE	ANNEX	BINDER	ABANDON
AXE	EXAM	EXTRA	NINTH	DINNER	BENEATH
BID	NEED	BUXOM	NOBLE	BAXTER	DENTIST
SIX	BURN	BOUND	EDITION	DEXTRIN	AMBITION

- a) ABANDON BURNED OXEN FOR DEAD.
- b) ONE MAN DID NOT BET.
- c) BAD ADDITION MADE NED DUMB.
- d) EXTRA BOXES ARE NEEDED IN MADISON.
- e) BOBS AMBITION MADE HIM BANAL.
- f) DON'T MENTION DRAB SUBJECTS AT DINNER.
- g) SEND NO NOTES ABOUT DISUNITED NATIONS.
- h) EXPANDED BASES NEED EXTRA MEN.
- i) INSANE INDIANS BEHEAD BUXOM DAMES.

LESSON SIX . . .

New characters used: G Q Z 7 K C Y

Exercises:

GUY	QUAY	GOOFY	GALLEY	KNUCKLE
COG	QUIZ	QUICK	QUARTZ	OXYGEN
QRA	AGED	CRAZY	KIBITZ	JONQUIL
DIZ	ZERO	KNACK	KINGLY	EXCUSE
KIN	KICK	CHECK	CAKCLE	13579
KEY	CODE	YACHT	PACIFY	GAZELLE
COY	COPY	QUACK	654720	73737
EKE	8714	96472	ZONITE	QUIRKS
719	1357	PYGMY	KAZOO	COCKY

- a) GOOFY PYGMY COGITATES QUIZZICALLY.
- b) JONQUILS GLORIFY KINGLY PACIFIC YACHTS.
- c) QUACKS CAN'T COPY CRAZY CODE GROUPS.
- d) KIBITZERS EXCUSE MY KNACK OF CACKLING.
- e) INQUIRING SCIENTISTS CHECK OXYGEN MASKS.
- f) PSYCHIC CRYSTAL GAZERS EKE CRAZY QUIRKS.
- g) AGED CYGNET HECKLES YOUNG GAZELLE.
- h) DIZZY QUIZZES QUICKLY GYRATE MINDS.
- i) GALLEY COOK SCRUBS DECKS WITH ZONITE.

At this point the instructor should conduct a general review of Lessons One through Six, including: additional practice in sending characters causing trouble to particular student; sending of plain text by students, instructor to comment on quality; sending of coded groups by students, instructor likewise to comment.

LESSON SEVEN . . .

New characters used: Common punctuation marks and procedure signals learned in a comparable receiving lesson.

Exercises:

In addition to sending of plain text with elementary punctuation included, students should send some messages in standard form, using all the necessary procedure signals; instructor to listen and comment, showing the correct method if any errors occur.

LESSON EIGHT . . .

Instruction from this point may take the form of lectures and exercises based on the "advanced" chapters of this booklet, covering high-speed operation, typewriter copy, "bug" sending, general operating data, etc.

SELECTIONS from the following text will be useful as practice material in tests for accuracy of copy. Speed of transmission should be somewhat below the average of the class, since this is exceedingly difficult text:

SENTIENT ENTENTE UTTER TRANSCONDUCTANCE EVASIVE JUJUTSU
KHANATE QUOIT OINTMENT MENTION ONION ONTOGENY REENTER
FAFNIR IRITIS TMESIS CENTENNIAL SENTIMENT SEDIMENT
MENDACIOUS OUSTER STERLING LINGERIE AERIAL ALUMINUM
NUMB TOMFOOLERY ERRONEOUSNESS ETESIAN JANISSARY
JINRIKISHA SHAH NIFLHEIM NUMBERED ZENANA NATANT
TANTAMOUNT ANTECEDE EDELWEISS XANTHIPPE PEPWERMINT

MISTRESSSHIP TETEATETE DEBILITATE PUNXSUTAWNEY MUTAUL
ZIGZAG YTTERBIUM MAMMA ENSIFORM XYST MNEMONIC ENFEOFF
BREVET XYLYLENE TSETSE BREE SHIRR ATAXIA HESSIAN
ATTESTATION OSSEOUS TATARIAN UITLANDER WHEEZE
CHLOROPHYLL DEVISEE MISSISSIPPI HOMONYM JINNEE ICHNEUMON
ZOOPHYTE LYDITE NAIVETE PYJAMA REREDOS QUEUE VIVIVOCE
GEISHA SKOVVEIEN ONOMATOPOEIA ULULATE YWAINE TIRESIAS
32343B SCHEHERAZADE ISIS FORESHEET XIPHODIAN MNEMOSYNE

TANTALUM UMLAUT AUTUMN MNEMONICS ICHTHYOPHAGOUS
NICTITATING NIHILISM TEUCRI ISEULT TENNESSEE JEREED
ORIENTATE JEQUIRITY SIESTA ARIES SIMOOM SIRENIAN APPANAGE
PATENTEE YTTRIA LAITY UDOMETER ERRANT VITEBSK ANTEATER
AORTA BACCARAT DEIFY COOPERAGE FANFARONADE TIENTSIN
GENRE HAWAIIAN QUARTERN KANAKA KAYAK LIGNIN NITER
DROHOBYCZ VERISIMILITUDE WAPITI VALLETTA CACATECAS
7M3432 LAUTTASAARI ENZYME MELLIFLUENCE BYSSUS VALUVULAR



**You'll Find it
in a
PUBLICATION**

The whole picture of amateur radio from basic fundamentals through the most complex uses of this appealing hobby is covered in ARRL publications. Old-time amateur, student or engineer, League publications

The whole picture of amateur radio fundamentals through the most complex hobby is covered in ARRL publications. Whether you are a beginner or an experienced ham, the ARRL publications will help you understand the ever-expanding field of electronics.

THE RADIO AMATEUR'S HANDBOOK

THE RADIO AMATEUR'S HANDBOOK
Internationally recognized, universally consulted. The all-purpose volume of radio. Packed with information useful to the amateur and professional alike. Written in a clear, concise manner. Contains hundreds of photos, diagrams, charts and tables. **\$6.00**

TUNE IN THE WORLD...WITH HAM RADIO

The complete beginner's package. Everything needed to obtain a Novice license: Theory, rules, how to assemble a station, and operating practices. With one-hour code cassette and free U.S. call-district wall map. **\$7.00**

HINTS AND KINKS

HINTS AND KINKS
No builder should be without this collection of practical ideas for the workshop and station. Gathered together from the "Hints and Kinks" column of QST. **\$2.00**

ARRL HAM RADIO OPERATING GUIDE

Written for the amateur who prides himself on good operating procedures. A ready reference source and guide to good operating practices. Ideal for the amateur who wishes to brush up on operating procedures and who wishes information on all facets of amateur operating. \$4.00

THE ARRL ANTENNA BOOK

THE ARKLE ANTENNA BOOK
Theoretical explanation and complete instructions for building different types of antennas for amateur work. Simple doublets, multielement arrays, mobile types, rotaries, long wires, rhombics and others. Transmission lines are exhaustively discussed. Profusely illustrated. \$4.00

LEARNING THE RADIOTELEGRAPH CODE

LEARNING THE RADIOTELEGRAPH CODE
For those who find it difficult to master the code. Designed to help the beginner. Contains practice material for home study and classroom use. **\$1.50**

THE RADIO AMATEUR'S VHF MANUAL

A thorough treatment of vhf. Covers receiving and transmitting principles, techniques and construction, antenna and feed system design, uhf and microwaves, test equipment, interference causes and cures. Emphasis throughout is on tried and tested equipment and practice.

A COURSE IN RADIO FUNDAMENTALS

A new book written to stand by itself. Twenty-six chapters present the electrical and electronic principles that are basic to understanding radio circuit operation. **\$3.00**

THE RADIO AMATEUR'S LICENSE MANUAL

THE RADIOAMATEUR'S LICENSE MANUAL
Study guide and reference book. Points the way toward the coveted amateur license. Complete with typical questions and answers to all of the FCC amateur exams — Novice, Technician, Conditional, General, Advanced and Extra Class. Continuously kept up to date. **\$1.50**

SINGLE SIDEBAND FOR THE RADIO AMATEUR

SINGLE SIDEBAND FOR THE RADIO AMATEUR
A digest of the best articles from QST. Includes discussions of theory and practical how-to-build-it descriptions of equipment. Covers reception and transmission. **\$3.00**

ARRL ELECTRONICS DATA BOOK

Includes data on radio frequency circuits, filter design, L, C, and R networks, antennas and feed systems. A catalog of solid-state circuits and much more. This 128-page reference is a useful addition to the amateur's technical library. \$4.00

SPECIALIZED COMMUNICATIONS TECHNIQUES

Seven chapters cover the more esoteric forms of amateur radio: ATV, SSTV, FAX, RTTY, Satellite Communications, and advanced communication techniques. **\$3.00**

UNDERSTANDING AMATEUR RADIO

FOR THE BEGINNER. Explains in simple language the elementary principles of electronic and radio circuits. Includes how-to-build-it information on low-cost receivers, transmitters and antennas. A "must" guide for the newcomer. **\$4.00**

FM AND REPEATERS FOR THE RADIO

FM AND REPEATERS FOR THE RADIO AMATEUR
A complete treatment of fm and repeaters. Includes chapters on receivers, transmitters, antennas as well as repeater operation. Of great value to the fm buff as well as to those just becoming interested in the subject. **\$4.00**